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REVIEW

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Overcoming resistance to antibody-drug conjugates: from mechanistic insights to cutting-edge strategies

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Abstract

Antibody-drug conjugates (ADCs) have revolutionized cancer therapy, but therapeutic resistance poses a significant barrier to sustained efficacy. Multiple mechanisms contribute to ADCs resistance, including drug efflux mediated by transporters, alterations in target antigens, tumor heterogeneity, and the impact of the tumor microenvironment (TME). Clinically, ADCs resistance varies across different cancer types, treatment lines, and patient subgroups. Emerging strategies now emphasize precision targeting through bispecific ADCs, alongside advancements in linker chemistry, payload design, and TME modulation. Additionally, rational combination therapies have emerged as a promising approach to reverse ADCs resistance, demonstrating synergistic effects. This review summarizes current understanding of mechanisms driving ADCs resistance and recent advances in combination therapies. By integrating mechanistic insights with emerging strategies, we aim to provide a comprehensive framework for addressing ADCs resistance and propose future research directions. These efforts may improve the efficacy of ADCs and the outcomes of cancer patients.

Keywords Antibody-Drug conjugates, Cancer, Resistance mechanism, Clinical manifestations, Innovative strategies, Tumor microenvironment

Background

Cancer remains one of the most significant global health challenges, with an estimated 20 million new cases and 9.7 million cancer-related deaths reported in 2022 [1]. This underscores the urgent need for more effective and precise treatment strategies. Despite progress in chemotherapy, targeted therapy, and immunotherapy, each treatment approach faces fundamental constraints [2–5]. Chemotherapy is hampered by poor selectivity, causing significant collateral toxicity [6–9], whereas targeted therapies and immune checkpoint inhibitors (ICIs) offer improved specificity but often with diminished direct tumor-killing potency [10, 11]. Thus, there is a critical demand for therapies that can specifically target tumor cells while minimizing collateral damage to healthy tissues [12].

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Antibody-drug conjugates (ADCs) enhance the therapeutic index through selective targeting of tumor-specific antigens, enabling precise cytotoxic drug delivery and mediating bystander killing effects against adjacent antigen-negative cancer cells [13–16]. Unlike chemotherapy, ADCs deliver cytotoxic payloads selectively to tumors via antibody targeting (detailed in Sect. "Comprehensive overview of ADCs"). This dual mechanism enhances efficacy while minimizing off-target toxicity, making them a transformative precision oncology platform [17]. Since their first FDA approval in 2000 for *gemtuzumab ozogamicin* (GO) in CD33-positive acute myeloid leukemia (AML) [18], several ADCs have been approved for the treatment of hematological and solid tumors. For example, *brentuximab vedotin* (BV) redefined treatment standards for relapsed/refractory Hodgkin lymphoma (HL) and systemic anaplastic large cell lymphoma (sALCL) through selective CD30-directed cytotoxicity [19], while *trastuzumab emtansine* (T-DM1) established a new cornerstone therapy for HER2-overexpressing breast cancer (BC) [20, 21]. The ADC landscape continues to advance at an accelerated pace, with an extensive pipeline of novel candidates currently undergoing evaluation. A comprehensive summary of currently approved ADCs was listed in Table 1, including their targets, evidence-based indications, and approved organizations [18–41].

Despite substantial progress in ADCs, their therapeutic potential remains constrained by several limitations, including dose-limiting toxicity, linker instability, and antibody immunogenicity. Among these challenges, acquired resistance poses the most critical barrier, undermining durable treatment responses through multiple mechanisms, such as antigen alteration, drug efflux via transporters, and tumor heterogeneity [42]. To overcome these hurdles, recent research has focused on developing precision strategies including next-generation targeting modalities, optimized antibody engineering approaches, tumor microenvironment (TME) modulation, and rational combination therapies. As illustrated in Fig. 1, these emerging solutions are advancing ADC design and clinical implementation, ultimately expanding their utility across diverse malignancies. This review provides a comprehensive analysis of ADC resistance mechanisms and explores innovative strategies to circumvent these challenges, thereby enhancing future clinical success and broadening their therapeutic applicability. Literature search encompassed peer-reviewed publications indexed in PubMed/Web of Science, official drug approval documents, and presentations from key oncology conferences.

Comprehensive overview of ADCs

ADCs revolutionize cancer therapy by precisely delivering cytotoxic drugs to tumors through antibody targeting. This smart design maximizes cancer cell destruction while sparing healthy tissue. While promising, their clinical success depends on addressing key challenges like drug resistance through mechanistic insights.

Structural composition and functional mechanism of ADCs

ADCs are composed of three main components: antibody, linker, and cytotoxic payload. The structures of ADCs were illustrated in Fig. 2A. The antibody provides specificity by selectively targeting tumor antigens, thus minimizing the impact on normal tissues and optimizing the delivery of the cytotoxic agent directly to cancer cells [43, 44].

Serving as a dynamic molecular scaffold, the linker enables spatiotemporal control of payload delivery. Cleavable designs exploit TME-specific triggers (acidic pH, cathepsins, or other enzymes), whereas non-cleavable linkers depend on lysosomal processing to release the cytotoxic agent [45, 46].

Cytotoxic payloads are ultra-potent drugs (e.g., microtubule inhibitors or DNA-damaging agents) that disrupt vital cellular functions. Their mechanism enables efficacy against both dividing and dormant tumor cells, ensuring wide therapeutic coverage [47, 48].

From mechanism to resistance

ADCs exert their therapeutic effects through a precise, stepwise mechanism. After binding to tumor antigens via the antibody, ADCs are internalized through receptor-mediated endocytosis and transported to lysosomes, where the cytotoxic payload is released. This release can be mediated by cleavable linkers, which respond to lysosomal conditions, or by non-cleavable linkers, which rely on the degradation of the entire ADC complex to release the payload. Once released, the payload induces apoptosis by either damaging DNA or disrupting microtubule dynamics [43]. In some cases, this can trigger an immune response via immunogenic cell death (ICD) [49]. Another feature of ADCs is the 'bystander effect', wherein membrane-permeable cytotoxic agents released from lysed target cells diffuse to neighboring antigen-negative cells, inducing cytotoxicity [50]. Figure 2B illustrates the mechanism of ADCs.

Although ADCs are designed to deliver cytotoxic payloads selectively to tumor cells while minimizing systemic toxicity, yet clinical efficacy is often limited by pharmacological challenges. Key issues include suboptimal antigen selection, inefficient lysosomal trafficking and premature linker cleavage in circulation. For instance, studies with the EGFR-targeting ADC Ab033 revealed that ~ 45% of internalized conjugates in A431 and H441 cell lines undergo cell surface recycling, delaying cytotoxic drug liberation and reducing intratumoral drug concentrations [51]. Additionally, unstable linkers can lead to premature payload release in the bloodstream, exacerbating

Table 1 Approved antibody-drug conjugates in cancer

Agent	Target	Approved indications	Approved by	Ref.
Gemtuzumab Ozogamicin	CD33	Newly-diagnosed CD33-positive AML	FDA, MFDS, NoMA, MHRA, PMDA, EMA, Swissmedic, TGA	[18]
Brentuximab Vedotin	CD30	Relapsed or refractory CD30 positive HL and sALCL	FDA, MFDS, NoMA, NMPA, MHRA, PMDA, EMA, Swissmedic, TGA	[19]
Trastuzumab Emtansine	HER2	1. Adjuvant treatment of patients with HER2-positive early BC who have residual invasive disease after neoadjuvant taxane and trastuzumab-based treatment 2. HER2-positive advanced BC	FDA, CDSCO, NMPA, NoMA, PMDA, EMA, Swissmedic, TGA	[20, 21]
Inotuzumab Ozogamicin	CD22	Relapsed or refractory CD22-positive B-cell precursor ALL	FDA, NMPA, EMA, PMDA, MHRA, Health Canada, TGA, MFDS, Swissmedic, NoMA	[22]
Moxetumomab Pasudotox	CD22	Relapsed or refractory HCL who have previously failed to receive at least two systemic therapies	FDA	[23]
Polatuzumab Vedotin	CD79B	Relapsed or refractory DLBCL	FDA, NMPA, EMA, PMDA, MHRA, Health Canada, TGA, MFDS, Swissmedic, TFDA	[24]
Enfortumab Vedotin	Nectin-4	Locally advanced or metastatic UC who have previously received platinum chemotherapy and a PD-L1/PD-1 inhibitor	FDA, NMPA, EMA, PMDA, MHRA, Health Canada, TGA, MFDS, Swissmedic, NoMA	[25]
Trastuzumab Deruxtecan	HER2	1. Unresectable or metastatic HER2-positive BC who have received two or more prior anti-HER2 based regimens in the metastatic setting; 2. Locally advanced or metastatic HER2-positive gastric or gastroesophageal junction adenocarcinoma who have received a prior trastuzumab-based regimen; 3. Unresectable or metastatic NSCLC whose tumors have activating HER2 mutations, and who have received a prior systemic therapy; 4. Previously treated HER2-low advanced BC	FDA, NMPA, EMA, PMDA, MHRA, Health Canada, TGA, MFDS, Swissmedic, NoMA	[26, 27, 28, 29]
Belantamab Mafodotin	BCMA	Relapsed or refractory MM in adult patients who have received at least four prior therapies, including an anti CD38 monoclonal antibody, a proteasome inhibitor, and an immunomodulatory agent.	FDA, EMA, PMDA, MHRA, Health Canada, Swissmedic, NoMA	[30]
Akalux	EGFR	Unresectable locally advanced or recurrent head and neck cancer	PMDA	[31
Loncastuximab Tesirine	CD19	Relapsed or refractory DLBCL	FDA, NMPA, EMA, Swissmedic, NoMA	[32]
Disitamab Vedotin	HER2	1. Previously treated (at least two systemic chemotherapy regimens) HER2-overexpressing (defined as IHC 2 + or 3+) locally advanced or metastatic GC (including gastroesophageal junction adenocarcinoma); 2. HER2-positive locally advanced or metastatic UC refractory to standard or regular therapies	NMPA	[33, 34]
Tisotumab Vedotin	Tissue factor	Recurrent or metastatic cervical cancer with disease progression on or after chemotherapy	FDA, EMA, PMDA, Swissmedic, NoMA	[35]
Mirvetuximab Soravtansine	FRα	Previously treated FRα positive, platinum-resistant epithelial ovarian, fallopian tube or primary peritoneal cancer	FDA, NMPA, EMA, MHRA, Swissmedic, NoMA	[36]
Sacituzumab Tirumotecan	Trop2	1. Previously treated metastatic triple-negative BC; 2. Previously treated advanced or metastatic nonsquamous NSCLC with EGFR mutations	NMPA	[37, 38]
Datopotamab Deruxtecan	Trop2	Hormone receptor positive, HER2 negative unresectable or recurrent BC after prior chemotherapy	FDA, NMPA, EMA, PMDA, Health Canada, TGA, Swissmedic, NoMA	[39]
Telisotuzumab Vedotin	cMET	Previously treated c-Met protein-overexpressing advanced nonsquamous EGFR-wildtype NSCLC	FDA	[40]
SHR-A1811	HER2	Previously treated, locally advanced or metastatic NSCLC with an activating HER2 mutation	NMPA	[41]

AML, acute myeloid leukaemia; HL, hodgkin lymphoma; sALCL, systemic anaplastic large cell lymphoma; BC, breast cancer; ALL, acute lymphoblastic leukaemia; HCL, hairy cell leukaemia; DLBCL, diffuse large B-cell lymphoma; UC, urothelial cancer; NSCLC, non-small cell lung cancer; MM, multiple myeloma; GC, gastric cancer; FDA, Food and Drug Administration (the United States); MFDS, Ministry of Food and Drug Safety (Republic of Korea); NoMA, Norwegian Medicines Agency (The Kingdom of Norway); MHRA, Medicines and Healthcare products Regulatory Agency (the United Kingdom); PMDA, Pharmaceuticals and Medical Devices Agency (Japan); EMA, European Medicines Agency (European Union); Swissmedic (Switzerland); TGA, Therapeutic Goods Administration (Commonwealth of Australia); NMPA, National Medical Products Administration (the People's Republic of China); CDSCO, Central Drugs Standard Control Organization (The Republic of India); Health Canada (Canada); TFDA, Thai Food and Drug Administration (The Kingdom of Thailand)

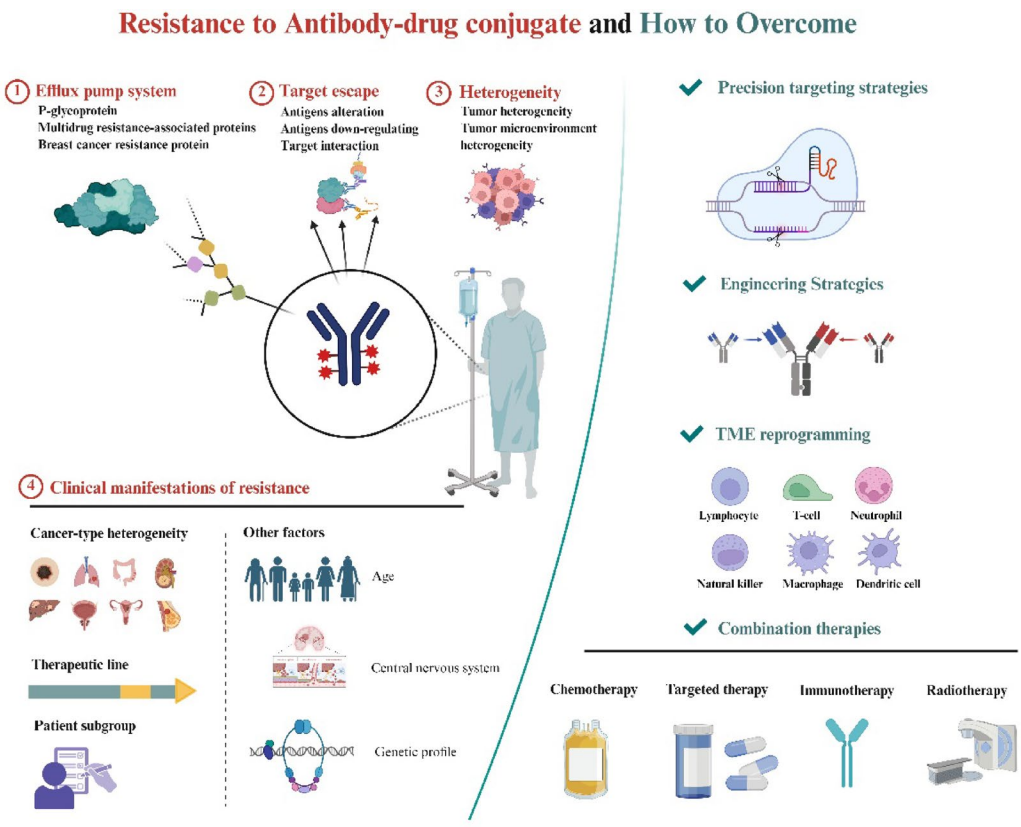

Fig. 1 An integrated illustration. The overview of key molecular mechanisms of ADCs resistance and their clinical manifestations. Emerging solutions, including next-generation ADCs technologies and synergistic strategies, to overcome resistance and enhance ADCs efficacy are summarized (Created with BioRender)

systemic toxicity while limiting effective drug delivery to tumors [52]. Addressing these limitations by implementing refined antigen targeting, strengthening lysosomal sorting mechanisms, and optimizing linker stability is essential to improving the efficacy of ADCs.

Mechanism of resistance

The emergence of resistance to ADCs is a complex process driven by dynamic interactions between various molecular mechanisms. These include antigen downregulation, changes in intracellular drug trafficking, and evasion strategies modulated by the immunosuppressive TME. Table 2 summarized the resistance mechanisms. A comprehensive understanding of these dynamic interactions is essential for developing next-generation ADCs with durable clinical responses (Fig. 3).

Efflux pump system

The efflux pump system, primarily comprising P-glycoprotein (P-gp), multidrug resistance-associated proteins (MRPs), and breast cancer resistance protein (BCRP), constitutes a central mechanism underlying resistance to ADCs [53]. These ATP-dependent transmembrane transporters actively export cytotoxic agents out of tumor cells, thereby diminishing intracellular drug accumulation and attenuating the efficacy of ADCs. Notably, efflux pumps cooperate with other resistance pathways to promote multidrug resistance (MDR), with each transporter subgroup contributing distinct but complementary drug extrusion capabilities.

P-glycoprotein

P-gp, encoded by the ABCB1 gene and classified within the ATP-binding cassette (ABC) transporter superfamily, is a well-established contributor to MDR in various malignancies [54]. Structurally, P-gp contains multiple transmembrane domains that create a central hydrophobic binding cavity. This cavity accommodates a wide range of structurally diverse substrates. When ATP is hydrolyzed, P-gp undergoes conformational changes, enabling energy-dependent drug efflux. Through this mechanism, it pumps anticancer agents from the cytoplasm back into the extracellular space, significantly reducing intracellular drug accumulation and diminishing therapeutic response [55, 56].

The payloads of ADCs are typically highly potent anti-cancer agents, such as auristatins and maytansinoids, which exert their therapeutic effects by targeting

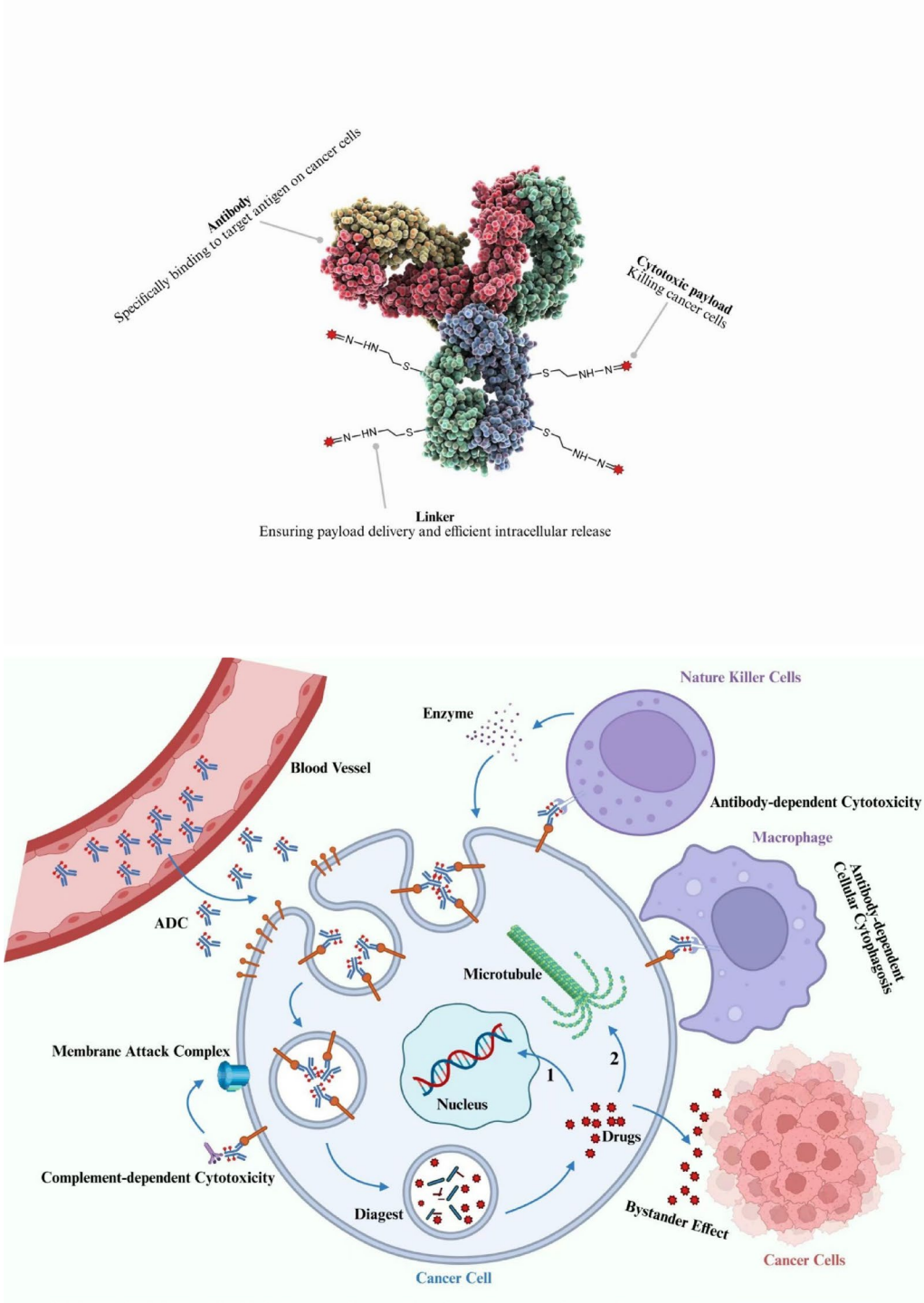

Fig. 2 Structure and mechanism of action of ADCs. **(A)** The structure of ADCs. **(B)** Mechanisms of actions of ADCs. ADCs, antibody-drug conjugates (Created with BioRender)

intracellular components like the microtubule system or DNA, thus inhibiting tumor cell proliferation or inducing apoptosis [57–60]. However, overexpression of P-gp remains a formidable obstacle to the clinical performance of ADCs. Several frequently used payloads, including calicheamicin gamma 1, monomethyl auristatin E (MMAE), DM1, and DM4, have been identified as P-gp substrates [61–64]. The diminished efficacy of GO and anti-Nectin-4-based ADC in P-gp-overexpressing tumor models further highlights the critical role of this efflux

Table 2 Resistance mechanisms of antibody-drug conjugates

Category	Resistance Mechanism	Overcoming Resistance
Efflux pumps system 1. P-glycoprotein 2. Multidrug resistance-associated proteins 3. Breast cancer resistance protein	Export cytotoxic agents out of tumor cells	1. Evade the recognition of efflux pumps system 2. Inhibit transporter 3. Suppresses the expression of transporter 4. Modulate metabolic pathway 5. Improve delivery system 6. Engineer payload
Target escape	1. Antigen mutations 2. Decrease expression 3. Aberrant target interactions	1. Monitor biomarkers 2. Regulate epigenetics 3. Discover target 4. Develop bispecific-targeting ADCs
Tumor heterogeneity	1. Harbor divergent driver mutations 2. Spatial heterogeneity 3. Interpatient heterogeneity	1. Dynamic monitor 2. Spatial proteomics analysis 3. Precise patient selection 4. Integrative multi-omics technologies
TME heterogeneity	1. Elevated interstitial fluid pressure 2. Intensive extracellular matrix 3. Activation of signaling pathways 4. Immunosuppressive cells 5. Hypoxia	1. Reduce intratumoral pressure 2. Improve vascular patency 3. Target fibroblast activation protein 4. Develop immune-stimulating antibody conjugates 5. Inhibit HIF-1α 6. Develop hypoxia-activated prodrugs
Clinical manifestations	1. Cancer-type heterogeneity 2. Therapeutic line 3. Patient subgroups 4. Age 5. Metastatic site	1. Precise patient selection 2. Personalized treatment 3. Molecular monitoring 4. Dose optimization 5. Combined therapy

ADCs, antibody-drug conjugates

pump in therapeutic failure [62, 65]. Furthermore, recent studies demonstrated that P-gp-mediated efflux of the DXd payload significantly impairs the cytotoxicity of *datopotamab deruxtecan* (Dato-DXd) in multiple cancer cell lines, supporting the notion that P-gp functions as a pan-ADC resistance determinant [66].

Clinically, high levels of P-gp expression have been correlated with poor responses to several approved ADCs. For example, in HER2-positive BC treated with T-DM1, the therapeutic synergy between *trastuzumab* and DM1 is compromised by P-gp-mediated DM1 efflux, leading to reduced treatment efficacy and disease progression [67]. Similarly, upregulation of P-gp in the BV-resistant HL cell line L428-R has been associated with acquired resistance [68]. These clinical and preclinical findings highlight the urgent need for strategies to circumvent P-gp-mediated efflux and restore ADC efficacy.

Several approaches are being explored to counteract P-gp-mediated resistance, focusing on disrupting its function, modulating its expression, or designing payloads with reduced P-gp affinity. One promising strategy involves rational payload engineering to develop cytotoxic compounds that evade P-gp recognition while retaining efficacy. *Exatecan*, a potent topoisomerase I inhibitor, is not a preferred substrate for P-gp and does not require metabolic activation. Building on this, Shia C-S et al. developed *OBI-992*, a TROP2-targeted ADC employing *exatecan* and an enzymatically cleavable linker, which demonstrated potent anti-tumor activity in vivo [69]. Importantly, *OBI-992* retained efficacy in P-gp-overexpressing cells, as *exatecan* showed markedly lower efflux compared to conventional payloads such as DXd and 7-ethyl-10-hydroxycamptothecin (SN-38) [66].

Alternatively, pharmacological inhibition of P-gp represents another approach. Third-generation P-gp inhibitors, including *tariquidar*, *zosuquidar*, *and laniquidar*, have shown potential in preclinical models to enhance intracellular drug retention by targeting the ATPase or substrate-binding domains of P-gp [70–73]. For instance, *tariquidar* was shown to restore drug sensitivity in anti-Nectin-4-resistant BC models with minimal systemic toxicity [62]. However, the clinical utility of systemic P-gp inhibitors remains constrained by their off-target effects on physiological barriers such as the blood-brain barrier (BBB), liver, and kidneys [54]. Beyond direct inhibition, targeting upstream signaling pathways regulating P-gp expression provides an indirect but potentially less toxic alternative. Activation of the PI3K/AKT and MAPK signaling cascades has been implicated in both tumor progression and P-gp upregulation [74–76]. Although this approach has yet to be applied specifically to ADCs, it offers a compelling rationale for future therapeutic modulation.

Gene-silencing strategies, particularly small interfering RNA (siRNA)-mediated targeting of P-gp, have demonstrated promising preclinical results in overcoming drug resistance, offering a more specific and potentially safer alternative to traditional chemical inhibitors with reduced systemic toxicity [77, 78]. In contrast to earlier-generation P-gp inhibitors, such as *verapamil and cyclosporine*, which required near-maximum tolerated doses but failed to substantially enhance efficacy due to their limited therapeutic windows and off-target effects [79]. siRNA-based silencing selectively suppresses P-gp expression with minimal interference with other transporters or metabolic pathways. This high level of specificity not only minimizes off-target toxicities but also provides sustained target inhibition, thereby overcoming the key pharmacological limitations of previous therapeutic strategies.

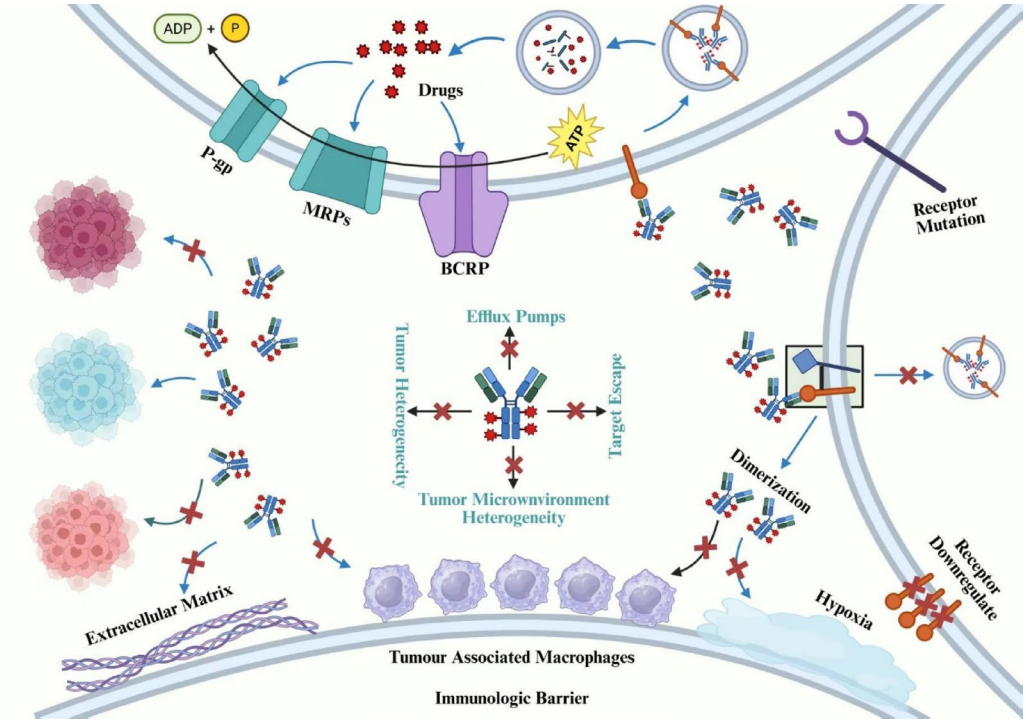

Fig. 3 Underlying mechanisms of resistance to ADCs. The resistance to ADCs are multifactorial and include efflux pump system, target escape, and tumor microenvironment heterogeneity. Efflux pumps system actively export cytotoxic agents from tumor cells, reducing intracellular drug accumulation. Tumor cells can also evade through antigen mutation, downregulation, or altered interactions with other cellular components. Additionally, the molecular heterogeneity within the tumor microenvironment, driven by subclonal evolution and microenvironmental selection pressures, further contributes to therapeutic resistance, limiting the efficacy of ADCs. ADCs, antibody-drug conjugates; P-gp, p-glycoprotein; MRPs, multidrug resistance-associated proteins; BCRP, breast cancer resistance protein (Created with BioRender)

However, siRNA delivery remains hindered by poor in vivo stability and immune activation [80], highlighting the need for optimized delivery platforms. Emerging technologies, including CRISPR/Cas9-based gene editing, offer additional strategies to suppress P-gp expression. Liposomal CRISPR/Cas9 systems have successfully downregulated P-gp and sensitized resistant cells to *doxorubicin* [81]. In parallel, nanoparticle-based delivery systems offer the dual benefit of protecting therapeutic payloads and modulating efflux activity. Nanocarriers have been shown to bypass or inhibit P-gp-mediated efflux, facilitating intracellular drug accumulation. Various platforms, including polymeric nanoparticles and liposomes, have been applied to encapsulate cytotoxic agents for MDR reversal [82, 83].

Although extensively researched, P-gp inhibition and drug delivery approaches remain unvalidated in the clinic for addressing ADC resistance. Given the broad substrate specificity and structural adaptability of P-gp, resistance mechanisms are often complex and context-dependent. Therefore, incorporating tumor-specific P-gp expression profiling into clinical decision-making may inform optimal ADC selection. Future efforts integrating structural biology, chemical pharmacology, and targeted delivery technologies are vital for developing effective solutions to P-gp-mediated resistance, ultimately enhancing therapeutic outcomes in ADC-treated patients.

Multidrug resistance-associated proteins

The MRPs, another key subfamily of ABC transporters, play a prominent role in mediating MDR across diverse cancer types [84]. Key members of the MRP family, including MRP1, MRP2, and MRP4, can export diverse xenobiotics and their conjugates through ATP hydrolysis-driven transport [85]. Despite variability in tissue distribution and substrate selectivity, MRPs commonly transport glutathione (GSH), sulfate, or GSH-conjugated molecules, underscoring their relevance in drug detoxification and resistance mechanisms [86].

MRP1 is one of the most extensively characterized members and is broadly expressed in various tumors [85]. It can export detoxified drug conjugates (e.g., GSH-adducted metabolites of chemotherapeutics), which lowers intracellular concentrations of bioactive drug forms and impairs ADC functionality [87]. In the case of T-DM1, for example, DM1 is released and forms GSH-conjugated metabolites, which are actively extruded by MRP1 [88, 89]. Knockdown of MRP1 has been shown to enhance sensitivity to *vincristine* and *etoposide*, while its overexpression confers resistance to these agents

[85]. Furthermore, elevated MRP1 expression has been observed in metastatic urothelial carcinoma (UC) lesions resistant to *enfortumab vedotin* (EV) [90].

MRP2 (ABCC2) is highly expressed in several solid tumors, including gastric, colorectal, and pancreatic cancers, and its upregulation has been linked to poor prognosis [91, 92]. Functionally, it exhibits preferential transport of organic anions and drug conjugates [93, 94]. SN-38, the bioactive metabolite of *irinotecan* and a common ADC payload compound, has been definitively identified as an MRP2 substrate. MRP2-mediated efflux has been implicated in resistance to SN-38 in pancreatic ductal adenocarcinoma (PDAC) models [92]. Moreover, MRP2 overexpression has been reported in *erlotinib*-resistant hepatoma cells [95] and T-DM1-resistant gastric cancer (GC) cells (N87-TDMR), where MRP2 inhibition restored drug sensitivity [96]. Similarly, upregulation of MRP2 was observed in cell lines resistant to pyrrolobenzodiazepine dimer-containing ADCs, and siRNA-mediated knockdown reversed resistance [97].

MRP4, another important efflux transporter, exhibits distinct substrate specificity compared to other MRP members, favoring the transport of nucleotide analogs and small molecules, such as prostaglandins and cyclic nucleotides [98, 99]. Emerging evidence suggests that MRP4 is closely associated with tumor proliferation. For example, Colavita JPM et al. reported MRP4 overexpression in clear cell renal cell carcinoma (ccRCC) compared to normal kidney tissue. Inhibition of MRP4 led to dysregulated cAMP metabolism, inducing cell cycle arrest, lipid composition changes, and apoptosis [100]. Similarly, PDAC with high MRP4 transcript levels exhibited more aggressive phenotypes, accelerated tumor progression, and resistance to *gemcitabine* [101, 102]. Moreover, MRP4 overexpression conferred resistance to topoisomerase I inhibitors, such as *irinotecan* and its active metabolite SN-38, in neuroblastoma models [103]. Although research on MRP4 in ADC resistance is limited, Liu-Kreyche P et al. employed targeted quantitative proteomics to identify MRP4 overexpression in cells resistant to BV and its payload, MMAE [61]. These findings suggest that MRP4 may play a role in ADC resistance, warranting further exploration to enhance ADC efficacy.

To address the resistance induced by MRPs, several strategies have been explored. A prominent approach involves the development of MRP inhibitors. *MK-571*, a classic MRP inhibitor, has been shown to block MRP1- and MRP4-mediated drug efflux, thereby enhancing the anti-tumor activity of various agents [104, 105]. For instance, *MK-571* induces apoptosis in colorectal cancer cells by activating MAPKs and caspase-3, while also inhibiting GSH-dependent efflux of benzyl isothiocyanate (BITC) to enhance its anti-proliferative effects [84]. In the context of ADCs, *MK-571* has been demonstrated to augment the cytotoxicity of GO, particularly in the presence of cyclosporine A (CSA) [106]. Additionally, pyrrolopyrimidine derivatives, such as *XR12890* and *XR13097*, exhibit potent MRP1 inhibition, reversing resistance to *doxorubicin* and *daunorubicin* in vitro and enhancing *vincristine*'s anti-tumor efficacy in lung cancer models [107]. Emerging MRP inhibitors, such as the deazapurine analog *ZW-1226*, have demonstrated superior efficacy compared to *MK-571*, effectively reversing MRP1-mediated resistance to topoisomerase II poisons, including *etoposide* and *doxorubicin* [108]. Research into multi-targeted ABC inhibitors, including *9-deazapurine*, *thienopyrimidine*, and *amino aryl ester*, has provided a foundation for developing inhibitors with broader specificity [85]. Despite these advancements, challenges such as low affinity, poor specificity, and safety concerns still remain. Recent study elucidating the transport mechanism of human MRP5 have opened new avenues for designing selective inhibitors, offering potential therapeutic benefits in MRP5-mediated MDR [109]. Whether these inhibitors can reverse ADCs resistance requires further investigation.

Modulating metabolic pathways, particularly those involving GSH, also represents another promising strategy. MRP family members, including MRP1 and MRP2, function as broad-spectrum efflux pumps for anionic substrates, particularly GSH-conjugated drugs. GSH, a critical endogenous antioxidant, plays a pivotal role in cellular detoxification and drug metabolism. Given that MRP1 substrates often include GSH-drug conjugates, intracellular GSH levels directly influence MRP-mediated drug efflux [110]. Thus, modulating GSH synthesis represents a promising strategy to enhance ADCs efficacy. *Buthionine sulfoximine (BSO)*, a γ-glutamylcysteine synthetase inhibitor that blocks the rate-limiting step in GSH synthesis, effectively depletes intracellular GSH level [111]. By reducing GSH concentrations, BSO diminishes cellular antioxidant capacity and detoxification, thereby potentiating the cytotoxicity of chemotherapeutic agents. For example, BSO has been shown to enhance the efficacy of *cisplatin*, a drug whose resistance is often associated with MRP1 overexpression [112]. In the context of ADCs, reducing GSH levels may impair MRP1-mediated efflux of cytotoxic payloads, leading to increased intracellular accumulation and enhanced anti-tumor activity. While direct evidence supporting this hypothesis remains limited, the functional interplay between MRP transporters and GSH metabolism offers a strong mechanistic rationale for exploring BSO-ADC combinations in future studies.

Additionally, structural modifications of ADCs to evade MRP recognition represent another innovative strategy. For instance, cleavable-linked auristatin-based

ADCs have demonstrated the potential to overcome MRP1-mediated resistance to T-DM1 [87]. Additionally, advanced delivery systems, such as lipid-based or polymeric nanoparticles, have been employed to enhance ADC accumulation in tumor cells while bypassing MRP-mediated efflux [113–116]. For example, enzyme-responsive polymeric nanoparticles such as EGFR antibody-modified, human serum albumin shelled mesoporous silica carriers have been developed, enabling pH- and MMP-2-triggered DOX release with high tumor selectivity and cytotoxicity, thereby exemplifying the potential of nanoparticle-based ADC delivery systems [115]. Gene-editing technologies also offer a targeted approach to reversing drug resistance. For example, siRNA-mediated downregulation of MRP2 reverses *cisplatin* resistance in oral squamous cell carcinoma [117], while inhibition of MRP1 sensitizes tumor cells to *epirubicin* [118]. Although the application of these technologies in ADC therapy remains underexplored, they hold significant potential for overcoming MRP-mediated resistance.

In conclusion, MRP family members represent a major barrier to ADC efficacy due to their drug efflux function. Overcoming this challenge requires: (1) developing selective MRP inhibitors with improved safety; (2) optimizing ADC payload design to evade MRP-mediated resistance; (3) validating rational combination therapies in preclinical/clinical models. Prioritizing these strategies will enable more effective and durable ADC-based treatments.

Breast cancer resistance protein

The BCRP, also known as ATP-binding cassette subfamily G member 2 (ABCG2), is a critical member of the ABC transporter superfamily. It is predominantly expressed in tissues such as the intestines, liver, hematopoietic stem cells, and mammary glands. Structurally similar to P-gp and MRP1, BCRP functions as a drug efflux pump. Its activity is driven by ATP hydrolysis and is primarily involved in the transport of a wide variety of small molecule drugs, hormones, and their metabolites. The spectrum of substrates recognized by BCRP is extensive, encompassing chemotherapeutic agents, antibiotics, and endogenous molecules [119, 120].

Recent studies have highlighted that BCRP plays a significant role in mediating resistance to ADCs by actively pumping out cytotoxic payloads such as MMAE, SN-38, and DM1. This reduces intracellular drug accumulation and thereby diminishes ADC efficacy. Specifically, ABCG2 is implicated in ADC resistance across a range of tumor types. For instance, Takegawa N et al. showed that HER2-positive GC cells develop resistance to T-DM1 via ABCG2-mediated efflux of DM1. Importantly, dual inhibition of ABCC2 and ABCG2 using *MK-571* restored T-DM1 sensitivity [96].

Addressing ABCG2-mediated resistance is crucial in overcoming therapeutic challenges associated with ADCs. The development of potent ABCG2 inhibitors has emerged as a pivotal strategy. One such inhibitor, *Ko143*, has shown promising preclinical efficacy in blocking ABCG2-mediated drug efflux. In medulloblastoma models, inhibition of ABCG2 by *Ko143* significantly enhanced the anti-tumor effects of *topotecan*, demonstrating the potential of ABCG2 inhibition in sensitizing tumors to chemotherapy [121]. Furthermore, a novel treatment approach, photodynamic therapy (PDT) with 5-aminolevulinic acid, has been explored for malignant brain tumors. In this context, glioblastoma cells expressing high levels of ABCG2 required higher light doses due to reduced accumulation of the photosensitizer. Notably, inhibition of ABCG2 with *Ko143* led to enhanced protoporphyrin IX (PpIX) accumulation and improved PDT efficacy [122]. Despite its potential, the metabolic instability of *Ko143* has prompted research into more stable analogs. For example, Zhu J et al. substituted *Ko143*'s labile tert-butyl ester group with an amide moiety, producing metabolically stable ABCG2 inhibitors with favorable pharmacokinetic properties in vivo [123]. In addition to synthetic inhibitors, natural compounds such as *resveratrol* have also shown promise in modulating BCRP activity, although further studies are required to establish their clinical applicability [124, 125].

Emerging evidence indicates that microRNAs (miRNAs) regulate BCRP expression, thereby modulating the pharmacokinetics of its substrate drugs. For example, miR-328-3p has been shown to target the 3'-untranslated region (3'-UTR) of BCRP mRNA, effectively regulating its expression and restoring chemosensitivity to *mitoxantrone* in MCF7 cells [126]. Similarly, miR-302a, by targeting the 3'-UTR of BCRP, reduces BCRP expression and enhances the sensitivity of BC cells to *mitoxantrone* [127]. These findings position miRNAs as promising candidates for overcoming BCRP-mediated resistance. However, the application of miRNA-based strategies in reversing ADC resistance is still in its early stages and requires further investigation.

In summary, as a member of the ABC transporter family, BCRP plays a critical role in mediating resistance to ADCs by extruding cytotoxic payloads, thereby limiting therapeutic efficacy. Strategies aimed at overcoming BCRP-mediated resistance, such as the development and optimization of specific inhibitors, as well as the application of gene-editing technologies, hold great promise for enhancing ADC-based therapies and overcoming resistance challenges.

Target escape

The therapeutic efficacy of antibody-drug conjugates (ADCs) is primarily governed by the high-affinity binding of monoclonal antibodies to tumor-specific antigens, enabling the precise delivery of cytotoxic payloads to malignant cells [128]. Despite targeted binding, tumor cells often evolve escape mechanisms that reduce ADC efficacy. These mechanisms include antigen mutations, downregulation of antigen expression, and interactions between the targeted antigen and other cellular components. Collectively, these factors contribute to reduced therapeutic responses, presenting significant challenges in clinical management [129].

A primary mechanism underlying resistance to ADCs involves genetic alterations of the target antigens. For example, *sacituzumab govitecan* (SG), the first FDA-approved ADC for the treatment of triple-negative breast cancer (TNBC), targets the trophoblast cell-surface antigen 2 (TROP2) and delivers a cytotoxic topoisomerase I inhibitor payload [130]. Despite its clinical success, resistance to SG has been increasingly reported. In a pivotal study, Coates et al. identified key genetic alterations contributing to SG resistance, including frameshift mutations in the TOP1 gene and a specific missense mutation in TROP2 (T256R). These mutations disrupt the trafficking of the target protein to the plasma membrane and impair antibody binding affinity, ultimately diminishing the therapeutic efficacy of SG [131]. Similarly, resistance to *inotuzumab ozogamicin* (IO), a CD22-targeted ADC used in B-cell acute lymphoblastic leukemia (B-ALL), has been attributed to acquired mutations in the CD22 gene. Zhao et al. demonstrated that loss of critical epitopes and structural remodeling of the CD22 protein significantly reduce antigen recognition by the ADC, contributing to treatment failure in relapsed B-ALL patients [132]. Further supporting this, Ryland et al. identified a truncating mutation within exon 4 of CD22, which results in reduced protein expression on cell surface, effectively abrogating ADC-mediated cytotoxicity [133]. These studies underscore the importance of genetic integrity of the target antigen in maintaining ADC efficacy and highlight the need for continuous biomarker monitoring (e.g., next-generation sequencing for mutation detection, immunohistochemistry for antigen expression profiling) and adaptive therapeutic strategies to overcome antigen-driven resistance.

Another mechanism of resistance involves the downregulation of target antigen expression by tumor cells [134]. A notable example of this phenomenon is observed in the treatment of HL and ALCL, where CD30 serves as a crucial therapeutic target [135, 136]. Also, BV has demonstrated significant clinical efficacy in these malignancies, achieving an overall response rate of 75%, a complete response (CR) rate of 34%, and a median progression-free survival (PFS) of 9.3 months in patients with relapsed or refractory HL [137]. In relapsed or refractory ALCL patients, the response rate is even more promising, reaching 86% [138]. However, despite these favorable outcomes, tumor cells can acquire resistance over time through mutations or downregulation of CD30 expression. This downregulation enables the tumor cells to evade the cytotoxic effects of the ADC. The efficacy of ADCs is not solely reliant on the specific binding between the antibody and its target antigen but also on the efficient internalization of the ADC into the tumor cell. Mutations or structural changes in the target antigen that impair internalization can prevent the conjugated toxin from entering the cell, ultimately compromising therapeutic efficacy. In a study by Chen et al., ALCL cells resistant to BV exhibited a significant downregulation of CD30 expression. While the absence of CD30 is not the sole mechanism of resistance, it significantly contributes by weakening antibody–antigen binding and thereby reducing the internalization efficiency of the ADC [139].

Similarly, a decrease in HER2 expression has been implicated in acquired resistance to T-DM1 and other HER2-targeted therapies. This further highlights the critical importance of maintaining target antigen expression in sustaining ADC efficacy [140]. Additionally, in trastuzumab-maytansinoid ADC (TM-ADC) resistant JIMT1-TM cells, Loganzo et al. observed a reduction of HER2 expression. This was accompanied by the development of cross-resistance to other trastuzumab-ADC formulations, further complicating resistance mechanisms and undermining the effectiveness of targeted therapies [141]. Clinically, this suggests that switching to another ADC targeting the same antigen may not be sufficient to overcome resistance.

The interactions between different targets on the cell surface also play a pivotal role in influencing the efficacy of ADCs. For example, EGFR, as a dimerization partner for HER2, modulates HER2 trafficking via endocytosis, thereby affecting the uptake of T-DXd. Gupta A et al. demonstrated that increased EGFR expression enhances the formation of EGFR/HER2 heterodimers, which inhibits the internalization of T-DXd and reduces its therapeutic efficacy. This observation points to a potential mechanism of resistance to T-DXd. Remarkably, disrupting EGFR expression through monoclonal antibodies or stimulating EGFR endocytosis can restore the transport and anti-tumor activity of T-DXd in EGFR-overexpressing cancers [142]. Moreover, antigen dimerization with other cell surface receptors can also contribute to ADC resistance. For instance, NRG-1β, a known ligand, induces HER2/HER3 heterodimerization. In subpopulations of HER2-amplified BC cell lines, this process can attenuate the cytotoxic activity of T-DM1. However, this resistance can be overcome by adding *pertuzumab*,

which blocks HER2/HER3 dimerization and disrupts downstream signaling. The combination of T-DM1 with *pertuzumab* has demonstrated a synergistic effect in vitro and vivo, underlining the potential of targeting HER2/HER3 interactions to improve ADC efficacy [143]. The emerging evidence highlights the complexity of ADC resistance mechanisms and emphasizes the importance of addressing target interactions to enhance the therapeutic outcomes of ADC-based therapies.

In conclusion, target antigen escape, driven by mutations, downregulation, and aberrant target interactions, remains a formidable barrier to the success of ADCs. Understanding these resistance mechanisms is essential for the development of rational strategies aimed at improving therapeutic efficacy.

Heterogeneity

Tumor heterogeneity

Tumor ecosystems are characterized by profound molecular heterogeneity, with distinct subclonal populations harboring divergent driver mutations. These subclones evolve through adaptive pathways influenced by microenvironmental selection pressures, contributing to therapeutic resistance [144–148]. In addition to genomic alterations, malignant cells often employ sophisticated resistance mechanisms, such as adaptive signaling rewiring (e.g., activation of the NOTCH pathway) or the acquisition of secondary genetic mutations (e.g., NTRK fusions, MET amplifications). These mechanisms collectively form complex resistance networks that complicate treatment strategies [149–153].

This molecular heterogeneity manifests across multiple biological scales, from intralesional clonal dynamics to intermetastatic variations. For example, despite uniform HER2 overexpression, certain subclones have been shown to develop resistance to T-DM1 through the downregulation of Endophilin A2 (Endo A2), which impairs endocytosis [154]. Loss of EndoA2 impairs endocytosis and lysosomal transport of HER2-ADC, leading to insufficient intracellular DM1 release and reduced cytotoxic effects [155]. In another study, Sung and colleagues developed a T-DM1-resistant model (N87-TM) by cyclically exposing HER2-positive GC N87 cells to clinically relevant concentrations of T-DM1. Comparative proteomic analysis revealed that N87-TM cells internalized T-DM1 into aberrant caveolin-1 (CAV-1)-positive endosomal compartments, bypassing the canonical lysosomal pathways and disrupting the intracellular trafficking necessary for effective payload release. This alternative endocytic routing offers a mechanistic explanation for acquired T-DM1 resistance [156].

In addition to molecular resistance mechanisms, spatial heterogeneity in the distribution of target antigens plays a critical role in determining the efficacy of ADCs. Systematic profiling of CLDN18.2 expression across various malignancies has revealed a patchy expression pattern in GC, with strong immunoreactivity observed in foveolar epithelial cells and the bases of gastric glands, while expression is nearly absent in the glandular neck regions [157]. This compartmentalized expression highlights significant intratumoral spatial variability, which presents a fundamental challenge for CLDN18.2-targeted ADCs. Complementing these findings, Qi C et al. developed a 68Ga-labeled nanobody PET imaging approach (68Ga-NC-BCH) to assess in vivo CLDN18.2 distribution. In gastrointestinal cancer patients with moderate-to-high CLDN18.2 expression (IHC 2+/3+), tracer uptake was highest in liver metastases, followed by nodal and bone lesions [158]. Notably, both inter- and intralesional heterogeneity were observed within individual patients, demonstrating fluctuating target availability across metastatic sites. These results suggest that the therapeutic efficacy of CLDN18.2-targeted ADCs may be limited by nonuniform antigen distribution, with certain tumor regions exhibiting either complete antigen loss or altered polarity. The polarity change refers to the mislocalization of antigen such as CLDN18.2 from the apical membrane to other cellular compartments, thereby reducing its accessibility to ADC binding. Given that the success of ADC payload delivery depends on uniform target engagement, such heterogeneity could contribute to partial resistance by shielding tumor subclones from effective drug binding. Clinically, this suggests that the efficacy of ADC may need to rely on image-guided patient selection to screen for lesions with preserved antigen polarity or a multifocal dosing strategy to overcome spatial heterogeneity.

Interpatient heterogeneity also plays a profound role in the efficacy of ADCs. A striking example can be found in HER2-positive breast cancer, where approximately 15–20% of cases exhibit either heterogeneous HER2 distribution or borderline-low expression (IHC 1+/2+ with negative FISH). This biological variability can impair T-DM1 binding kinetics and cellular uptake, leading to suboptimal payload delivery. In these instances, some tumors that are pathologically classified as HER2-positive may exhibit functional resistance [159, 160]. These findings reveal a significant gap between conventional biomarker classifications and real-world drug-target interactions. This highlights the urgent need for advanced predictive approaches, including artificial intelligence (AI) powered quantitative histomorphometry and spatial proteomics analysis. Such next-generation frameworks should move beyond traditional IHC and FISH diagnostic thresholds to more accurately predict ADC treatment response.

To address the challenges associated with ADCs resistance, integrative multi-omics technologies, combined

with dynamic monitoring strategies, provide a promising framework for redefining therapeutic paradigms. Single-cell RNA sequencing (scRNA-seq), paired with spatial transcriptomics, enables precise mapping of clonal evolution and the identification of critical resistance-associated genes and pathways. A notable example of this approach is demonstrated by Izumi M et al., who utilized multi-omics strategies to investigate apoptosis-related transcriptional reprogramming during the acquisition of resistance to EGFR tyrosine kinase inhibitors (EGFR-TKIs) in EGFR-mutant lung cancer. Their study identified BCL2L1, which encodes the anti-apoptotic protein, as a central factor in driving therapeutic evasion through its selective upregulation in tumor cells [161]. This methodology has been successfully extended to predict resistance mechanisms not only to targeted therapies but also to immunotherapy and chemotherapy.

Furthermore, DNA damage response (DDR) mechanisms, which contribute to chemoresistance, are critical areas of investigation within ADC therapy. Sun et al. employed RNA-seq and scRNA-seq to uncover EGLN3, a pivotal gene involved in DDR, and assessed its spatially heterogeneous expression through spatial transcriptomics. Their findings revealed that elevated EGLN3 expression correlates with poor prognosis in lung adenocarcinoma, particularly in hypoxic and neoplastic regions. Functionally, EGLN3 was shown to promote resistance to cisplatin and docetaxel, while modulating TGF-β signaling, regulatory T cell (Treg) infiltration, and cancer-associated fibroblast (CAF) activation, thus contributing to resistance to ICIs [162, 163]. These insights into DDR mechanisms emphasize the utility of multi-omics approaches in understanding and overcoming resistance to various therapies, including ADCs.

While the application of multi-omics strategies to overcome ADC resistance is still an evolving field, its translational potential is already evident. For example, optimizing ADC therapy could involve sequential administration of T-DM1 to target HER2-high clones, followed by T-DXd to target low-expression clones. In addition, circulating tumor DNA (ctDNA) profiling offers a non-invasive means of monitoring clonal evolution and tracking emerging resistance mutations in real time. Kim et al. demonstrated the utility of ctDNA in predicting molecular subtype-based responses to *pembrolizumab* in GC, revealing differential responses across subtypes [164]. Similarly, Sanchez-Vega F et al. identified acquired MET amplification as a key driver of resistance to afatinib, a pan-HER kinase inhibitor, in HER2-positive esophagogastric cancer, using longitudinal ctDNA monitoring [165]. This finding suggests that early inhibition of c-MET could potentially preempt or reverse resistance to HER2-targeted therapies.

Collectively, these multi-omics approaches not only help mitigate intrinsic ADC resistance but also pave the way for the development of adaptive therapeutic regimens tailored to the evolving molecular landscape of tumors. As such, these strategies hold significant promise for advancing the field of precision oncology, particularly in the context of ADC-based treatments.

TME heterogeneity

The highly heterogeneous TME critically impairs ADC delivery and efficacy by driving multifaceted resistance mechanisms. Key among these barriers are elevated interstitial fluid pressure (IFP) and intensive extracellular matrix (ECM). Together, these structural features create formidable physical obstacles that severely limit ADC penetration and homogeneous intratumoral distribution. Notably, the accumulation of hyaluronan (HA) in pancreatic and breast carcinomas has been shown to substantially elevate IFP, disrupting drug penetration kinetics and altering biodistribution patterns [166, 167]. Mechanistically, the interaction between HA and its receptor CD44 activates pro-survival signaling pathways, including MAPK and PI3K/AKT, while also accelerating epithelial-mesenchymal transition (EMT). These processes foster tumor recurrence and contribute to therapeutic resistance [168]. Preclinical investigations utilizing PEGylated hyaluronidase (PEGPH20) successfully reduced intratumoral pressure and improved vascular patency, leading to enhanced therapeutic outcomes [169, 170]. Subsequent phase II clinical trials (NCT01839487) demonstrated improved PFS in patients with HA-high advanced PDAC treated with PEGPH20 in combination with *nab-paclitaxel* and *gemcitabine* [171]. However, a phase III trial (NCT02715804) failed to confirm survival benefits, highlighting the translational challenges of this strategy.

CAFs are crucial stromal components within the TME that orchestrate drug resistance by secreting growth factors and ECM proteins, which establish additional physical barriers to drug delivery [172]. Integrated analyses of gene expression datasets and whole-transcriptome sequencing have elucidated CAF-mediated resistance mechanisms in hepatocellular carcinoma (HCC), specifically through the secretion of SPP1. This CAF-derived factor activates parallel oncogenic pathways and induces EMT, thereby conferring resistance to TKIs, such as *sorafenib* and *lenvatinib* [173]. Furthermore, CAFs modulate matrix stiffness, further impeding drug diffusion. For instance, calponin 1 (CNN1)-mediated ECM stiffening was shown to confer resistance to *5-fluorouracil* in GC models, with CNN1 knockdown restoring chemosensitivity [174]. Although evidence for CAF-mediated ADC resistance remains limited, targeting CAFs represents a promising therapeutic strategy. For example,

ADCs targeting fibroblast activation protein (FAP) have shown significant progress. *OMTX705*, a novel FAP-targeting ADC, exhibited potent anti-tumor activity in PD-1-resistant models by enhancing CD8+ T cell infiltration [175]. Similarly, the ADC *huB12-MMAE* demonstrated specific cytotoxicity against FAP + cells in prostate cancer models [176]. Overall, CAFs contribute to tumor progression by both creating physical barriers, such as ECM stiffening, which impede drug diffusion, and promoting adaptive drug resistance in tumor cells through paracrine signaling mechanisms. This dual role of CAFs is a key factor limiting the efficacy of ADCs and underscores the rationale for targeting CAFs as a therapeutic strategy.

The immunosuppressive cells within the TME, such as tumor-associated macrophages (TAMs), establish formidable immunological barriers that not only restrict anti-tumor immune responses but may also impair the delivery and cytotoxic efficacy of ADCs. TAMs are well-recognized drivers of tumor progression, and increased infiltration has been correlated with poor clinical outcomes. Previous studies have shown that TAMs can reduce tumor cell sensitivity to chemotherapeutics, including *paclitaxel* and *gemcitabine*, through the suppression of CD8+ T cell activity [177, 178]. Paradoxically, while TAMs demonstrate pro-tumorigenic properties, they also retain Fc-dependent anti-tumor capabilities. The Fc–Fcγ receptor (FcγR) axis mediates both monoclonal antibody (mAb)-directed effector cell activity and antibody-dependent agonistic signaling via receptor clustering. Waight et al. demonstrated enhanced antigen-specific T cell responses through FcγR engagement on antigen-presenting cells (APCs) with anti-CTLA-4 and anti-TIGIT mAbs [179]. Similarly, the efficacy of ADCs fundamentally depends on Fc-FcγR interactions, as the Fc-mediated effector functions significantly contribute to tumor cell elimination. Preclinical studies have established the role of TAM-mediated FcγR interactions in ADC processing and drug release [180]. However, emerging research has identified potential safety concerns associated with ADC aggregates, which may inadvertently activate immune cells via FcγR engagement and internalization, potentially leading to off-target toxicity [181, 182]. The varied tissue distribution of FcγR + macrophages, especially their prevalence in the bone marrow, spleen, and liver, may contribute to inconsistent ADC biodistribution and unpredictable therapeutic outcomes in vivo [183]. To mitigate FcγR-mediated toxicity, Dimasi N et al. engineered site-specific cysteine insertions in ADC antibodies to create variants that abrogate FcγR binding in vitro, while retaining potent, dose-dependent anti-tumor activity in BC models [184]. In parallel, Guo Q et al. addressed toxicity limitations associated with *rovalpituzumab tesirine* (Rova-T) in small-cell lung cancer (SCLC) by developing *FZ-AD005*, a DLL3-targeted ADC incorporating Fc-silencing technology. Preclinical studies demonstrated its specific binding, efficient internalization, bystander cytotoxicity, and potent anti-tumor activity, without off-target toxicity [185]. These findings underscore the need for further investigations into the role of TAMs in ADCs efficacy and payload release, along with broader safety assessments.

Hypoxia, a hallmark feature of the TME, primarily results from aberrant vasculature and increased oxygen consumption by rapidly proliferating tumor cells [186]. This hypoxic environment not only impairs ADCs delivery efficiency but also reduces therapeutic efficacy through pharmacokinetic alterations and metabolic adaptations in tumor cells. For example, Indira Chandran V et al. demonstrated that hypoxia promotes resistance to T-DM1 in HER2-positive BC. Hypoxia-induced translocation of CAV-1 from cytoplasmic vesicles to the plasma membrane disrupts *trastuzumab*/HER2 complex internalization, driving resistance to treatment [187]. Additionally, hypoxia downregulates the expression of targeted antigens such as PD-L1, further contributing to ADCs resistance [188]. Hypoxia also rewires tumor metabolism, as exemplified by HIF-1α-mediated glycolytic reprogramming, which confers resistance to *5-fluorouracil* in colorectal cancer [189]. Notably, hypoxic conditions can induce resistance to *SN-38*, a commonly used payload in ADCs [190].

To address these challenges, emerging strategies are focused on modulating hypoxia to enhance drug efficacy, such as HIF-1α inhibition or the development of hypoxia-activated prodrugs [191–193]. Jayaprakash P et al. reported that the hypoxia-activated prodrug *TH-302*, when combined with ICIs, reduced myeloid-derived suppressor cell (MDSC) infiltration while enhancing T cell-mediated anti-tumor responses [193]. In an innovative approach, Wang Y et al. designed an oxygen-sensitive ADC that selectively releases MMAE under hypoxic conditions. This ADC demonstrated superior efficacy and safety profiles in HER2-positive BC and GC models compared to conventional ADCs [194]. Additionally, metabolic interventions hold promise for overcoming hypoxia-driven resistance. For instance, pyruvate dehydrogenase kinase (PDK) inhibition restored sorafenib-induced apoptosis sensitivity in sorafenib-resistant HCC models [195]. Although hypoxia remains a critical mediator of ADCs resistance, these advances offer novel avenues for therapeutic optimization, particularly through hypoxia-responsive ADCs in combination with metabolic modulators.

In conclusion, resistance to ADCs mediated by the TME involves complex biological pathways that contribute to treatment failure at multiple levels. Overcoming these barriers requires a systematic approach that combines structural refinements of ADCs (e.g., optimized

Fc engineering and hypoxia-responsive payload release) with novel therapeutic combinations (e.g., CAF-targeted agents or metabolic pathway modulators). Each of these strategies presents distinct opportunities to enhance ADC efficacy. Continued exploration of TME-ADC interactions will yield new approaches to circumvent resistance mechanisms, providing critically needed therapeutic alternatives for treatment-resistant cancers.

Clinical manifestations of resistance mechanisms

Preclinical studies have elucidated the key mechanisms of resistance to ADCs. However, clinical data from real-world settings reveal a more intricate landscape of ADCs resistance, characterized by pronounced tumor-type specificity and differential responses based on treatment lines and patient subgroups.

Cancer-type heterogeneity

The resistance profiles of ADCs exhibit significant variability across different cancer types. This is particularly evident in the context of HER2-targeting ADCs. T-DM1, for instance, has demonstrated robust efficacy in HER2-positive BC, with the phase III EMILIA trial confirming substantial improvements in both PFS and overall survival (OS) in patients with metastatic disease [196]. However, its clinical benefit differs substantially in other indications. The phase II/III GATSBY trial revealed that T-DM1 provided limited benefit for patients with previously treated HER2-positive locally advanced or metastatic gastric/gastroesophageal junction adenocarcinoma, achieving a mOS of 7.9 months and mPFS of 2.7 months, which failed to outperform taxane-based therapy [197].

Similarly, T-DXd demonstrated superior anti-tumor activity in *trastuzumab*-pretreated HER2-positive metastatic BC, with a mPFS of 16.4 months [198]. However, in the DESTINY-Gastric02 trial, T-DXd exhibited a significantly diminished efficacy in HER2-positive advanced GC, showing a mPFS of only 5.6 months in *trastuzumab*-pretreated patients [199]. This represents a clinically meaningful efficacy gap when compared to BC cohorts.

For HER2-positive non-small cell lung cancer (NSCLC), the efficacy of ADCs remains investigational. Preliminary data from the DESTINY-Lung01 trial indicated that T-DXd achieved a mPFS of 5.7 months and mOS of 12.4 months in patients with unresectable or metastatic NSCLC harboring HER2 overexpression or ERBB2 mutation [200]. However, these outcomes were still inferior to those observed in BC patients, likely due to the pronounced intratumoral heterogeneity of HER2 expression in NSCLC. These findings were further corroborated by the DESTINY-PanTumor02 study, which observed objective responses across all HER2-expressing (IHC 3+/2+) locally advanced/metastatic solid tumor cohorts but also revealed significant disparities in PFS. The endometrial cancer cohort had the longest mPFS (11.1 months), while the pancreatic and biliary tract cancer cohorts showed notably shorter PFS (3.2 and 4.6 months, respectively) [201]. Furthermore, the phase I/II IMMU-132-01 basket trial revealed considerable variability in the activity of SG, across different refractory metastatic epithelial cancers, with mOS ranging from 4.5 to 13.0 months and mPFS varying from 2.0 to 6.8 months [202]. The phase I trial of *RC48*, a novel HER2-targeted ADC, further emphasized the heterogeneity of ADC efficacy across malignancies. Subgroup analysis in 47 GC patients showed an objective response rate (ORR) of 21.3%, while all 4 UC patients achieved responses (ORR, 50%, disease control rate (DCR), 100%). In contrast, none of the six patients with other cancers (including breast, colorectal, and gallbladder cancer) showed partial response (PR) [203]. These findings prompted subsequent trials in gastric and urothelial cancers, which validated the efficacy of RC48 in these selected populations [204, 205].

These data clearly demonstrate that while target-specific ADCs have shown promise in clinical trials, substantial cancer-type heterogeneity remains a critical factor influencing treatment response. Some malignancies exhibit rapid acquired resistance and early disease progression, which are likely driven by variations in target antigen expression, TME characteristics, and tumor biology inherent to each cancer type.

Impact of therapeutic line on ADCs resistance

In addition to cancer-type differences, the efficacy of ADCs is significantly influenced by prior treatment lines. Resistance to ADCs becomes increasingly prevalent as patients progress through successive treatment lines. For instance, subgroup analysis from the EMILIA trial revealed a decline in T-DM1 efficacy in more advanced treatment settings [196]. Similarly, the phase III EV-301 trial, which evaluated EV in patients with locally advanced or metastatic UC who had previously received *platinum*-based chemotherapy and experienced disease progression during or after PD-1/PD-L1 inhibitor therapy, found that the survival benefit diminished with increasing lines of prior therapy. For patients who had received 1–2 prior lines of therapy, the hazard ratio (HR) was 0.69, whereas for those with ≥ 3 prior lines, the HR increased to 0.88 [206].

Similarly, SG demonstrated superior OS and PFS when administered as third-line therapy for metastatic TNBC compared to fourth-line or later [207]. In platinum-resistant ovarian cancer (OC), *mirvetuximab soravtansine* (MIRV), an ADC targeting FRα, reduced the risk of death by 49% (HR = 0.51) in the second-line setting, whereas its efficacy was notably diminished in subsequent-line treatments [208].

These data clearly demonstrate that the efficacy of ADCs is significantly influenced by the line of treatment. Whether the frontline therapies induces ADC resistance or whether there is cross-resistance remains to be further investigated in future studies.

Efficacy across patient subgroups

The efficacy of ADCs also varies significantly across different patient subgroups. In the phase III TROPION-Lung01 trial, Dato-DXd significantly improved survival outcomes compared to *docetaxel* in pretreated advanced/metastatic NSCLC. However, subgroup analyses revealed enhanced clinical benefit in patients with non-squamous histology (mPFS of 5.6 vs. 3.7 months) and those harboring actionable driver mutations (ORR of 26.3% vs. 12.3%) [209]. Similarly, the EVOKE-01 study evaluating SG reported comparable efficacy between squamous and non-squamous NSCLC cohorts. However, stratification by prior ICI response status revealed critical divergence in therapeutic benefit. SG conferred superior mOS (11.8 months; HR = 0.75) in ICI-refractory patients, whereas minimal survival advantage was observed in ICI responders (HR = 1.09) [210]. This dichotomous response underscores the influence of prior immunotherapy on ADCs efficacy.

This pattern of response heterogeneity also extends to breast cancer. In metastatic BC, T-DXd demonstrated sustained anti-tumor activity, with the DESTINY-Breast01 trial reporting an ORR of 60.9% in HER2-positive cases [211]. In the follow-up DESTINY-Breast04 trial in HER2-low metastatic BC (IHC 1 + or 2+/ISH-), T-DXd showed superior efficacy compared to chemotherapy, with significant improvements in PFS (9.9 vs. 5.1 months, $P < 0.001$) and OS (23.4 vs. 16.8 months, $P = 0.001$) [212]. However, these benefits were less pronounced than in HER2-positive cases (mPFS of 17.8 months; mOS of 39.2 months), illustrating a clear gradient of efficacy based on HER2 expression. Likewise, in HER2-positive GC (defined by IHC 3 + or IHC 2+/ISH+), the DESTINY-Gastric01 study found that T-DXd significantly prolonged mOS (12.5 months) [213]. However, the survival advantage weakened in HER2-low subgroups, with mOS reducing to 7.8 months (IHC 2+/ISH-) and 8.5 months (IHC 1+) [214]. This consistent HER2-dependent efficacy pattern across different malignancies highlights the critical role of HER2 expression as a biomarker for ADCs response, reinforcing the importance of precise patient stratification in clinical practice.

Other factors

Resistance to ADCs in hematologic malignancies often exhibits a pronounced age-dependent pattern. Elderly patients, particularly those over 75 or with compromised performance status, may demonstrate reduced tolerance to conventional therapies. In a pivotal study evaluating GO in AML patients aged >60 years, the CR rate was 33% for patients aged 61–75, compared to only 5% in those aged >75. Moreover, mOS was significantly longer in the 61–75 cohort ($P = 0.05$) and among CD33 + cases ($P = 0.05$) [215], highlighting both age and antigen density as crucial determinants of ADC efficacy.

Therapeutic distribution of ADCs can also be compromised by sanctuary sites such as the central nervous system (CNS). T-DXd demonstrated diminished efficacy in HER2-positive BC patients with CNS metastases, evidenced by a lower ORR (51.7%) versus extracranial disease (62.7%) [216]. This disparity underscores the BBB as a significant impediment to ADC penetration, contributing to local therapeutic failure in CNS-involved malignancies.

The current clinical evidence highlights the complexity of ADC resistance, emphasizing the need for a more integrated therapeutic approach. Central to this strategy is the implementation of personalized treatment tailored to individualized tumor biology and genetic profile. In advanced-line therapy, careful sequencing is critical to avoid redundant use of agents with overlapping mechanisms. Furthermore, continuous molecular monitoring, such as single-cell sequencing or circulating tumor cells (CTCs), will provide valuable insights into clonal evolution and the emergence of resistant subpopulations. Special attention must also be given to optimizing dosing regimens for vulnerable patient groups, such as the elderly, to achieve optimal therapeutic efficacy while minimizing treatment-related toxicities.

Strategies to overcome ADCs resistance

To address these challenges, researchers have pioneered multipronged precision therapeutic strategies, including target-specific modulation, structure-guided antibody engineering, rational nanocarrier design, and spatiotemporal TME reprogramming. Here, we critically analyze these advances, with a focus on their mechanistic rationale and clinical translational potential (Fig. 4).

Precision targeting strategies

The primary driver of drug resistance involves target-related aberrations, underscoring the need to enhance the target expression stability or broaden the modality of therapies.

Epigenetic regulation

Epigenetic modulation represents a promising strategy to overcome ADCs resistance by restoring or enhancing tumor antigen expression through mechanisms such as DNA methylation and histone modification [217, 218].

For instance, *decitabine* (DAC), a DNMT inhibitor, is directly incorporated into DNA, leading to suppression

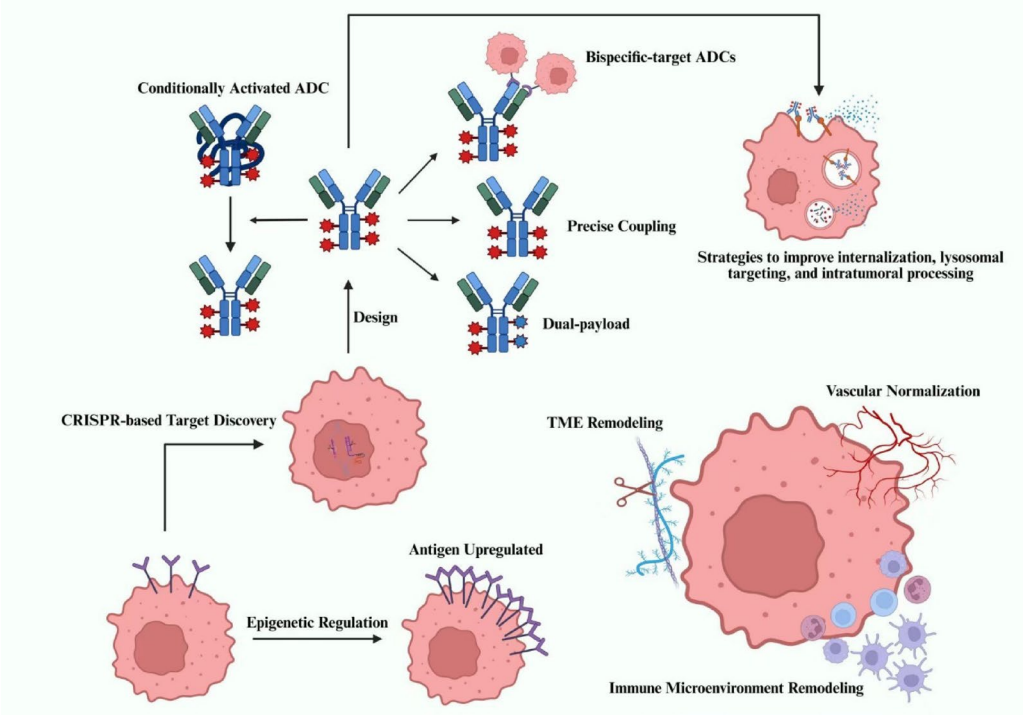

Fig. 4 Strategies to overcome ADCs resistance. Strategies to overcome resistance to ADCs focus on enhancing target stability, improving payload delivery, and addressing TME challenges. Epigenetic modulation, such as DNA methylation and histone modification, can restore tumor antigen expression. Dual or multi-target ADCs, by targeting multiple antigens, improve specificity and payload internalization. Advances in payload engineering aim to overcome efflux pump-mediated drug clearance and lysosomal sequestration. Moreover, reprogramming the TME through targeting both cellular and extracellular components is crucial to enhance ADC efficacy. These strategies collectively address resistance mechanisms and improve the potency and selectivity of next-generation ADCs. ADCs, antibody-drug conjugates; TME, tumor microenvironment (Created with BioRender)

of DNA methyltransferase activity [219]. Compelling evidence demonstrates that DAC could upregulate the presentation of tumor antigen via demethylation. A study performed by Ma R et al. demonstrated that DAC-driven DNA hypomethylation epigenetically reactivates neoantigen-encoding transcripts and cancer-testis antigens (CTAs) in glioblastoma, thereby augmenting MHC class I-dependent antigen presentation and priming tumor-reactive T cell responses, ultimately enhancing immunosurveillance [220]. This insight has been further validated in head and neck squamous cell carcinoma (HNSCC) [221]. Moreover, in melanoma, DAC synergized with ICAM1-targeted ADCs to suppress tumor growth both in vitro and in vivo, without significant off-target toxicity. This effect was attributed to DAC-induced upregulation of ICAM1 expression and enhanced ADCs internalization, thereby potentiating cytotoxic payload delivery and apoptosis [222]. In BC, Zhao et al. demonstrated that DAC treatment downregulated DNMT levels and demethylated the TROP2 promoter, leading to robust surface TROP2 upregulation. This not only improved the binding efficiency of SG but also enhanced tumor cell killing [223].

Beyond tumor antigen restoration, DAC exerts pleiotropic effects on the TME. In colorectal cancer models, DAC markedly upregulated PD-L1 expression [224], while in TP53-mutant diffuse large B-cell lymphoma (DLBCL), it inhibited SUV39H1-mediated H3K9me3 deposition at endogenous retroviral elements, triggering a type I interferon response and enhancing CD4+/CD8 + T cell activation [225]. Notably, Li X et al. showed that DAC reactivates and expands intratumoral CD8 + progenitor exhausted T cells, thereby improving responses to anti-PD-1 therapy. Compared to monotherapy, the DAC-anti-PD-1 combination significantly suppressed tumor growth in multiple preclinical models [226]. Collectively, these findings have catalyzed numerous clinical investigations of DAC-based combinatorial regimens, particularly in hematological malignancies, with encouraging outcomes reported to date [227–232].

Enhancer of zeste homolog 2 (EZH2), the catalytic subunit of polycomb repressive complex 2 (PRC2), mediates transcriptional silencing via trimethylation of histone H3 at lysine 27 (H3K27me3). Pharmacological inhibition of EZH2 reverses this epigenetic repression and enhances antigenicity. In HNSCC, EZH2 inhibitors upregulated MHC class I expression, increased CD8 + T cell infiltration, and restored sensitivity to anti-PD-1 therapy in murine models [233, 234]. In hematologic contexts, *tazemetostat* synergized with IFN-γ to elevate MHC class I

expression in DLBCL [235]. Moreover, *valemetostat*, a dual EZH1/2 inhibitor, reversed CD19 CAR-T resistance by reactivating B-cell signaling pathways and proinflammatory gene networks [236]. Although preclinical and clinical data directly associating EZH2 inhibition with improved ADC efficacy are still emerging, ongoing clinical trials (e.g., NCT05994235) are currently investigating potential synergistic effects between EZH2 inhibitors and ADCs in hematologic malignancies.

Histone deacetylase inhibitors (HDACis) also reverse epigenetic silencing and have been shown to rescue antigen expression, primarily by enhancing MHC class I-mediated presentation and promoting immune effector infiltration [237–241]. Preclinical study indicated potent synergy between *vorinostat* and *camidanlumab* tesirine, a CD25-targeted ADC, in T-cell lymphoma [242]. Early-phase clinical trials combining HDACis and GO in AML patients have demonstrated safety and preliminary efficacy [243, 244]. However, larger randomized trials are needed for conclusive validation.

CRISPR-Based target discovery

The refinement of genome-wide loss-of-function screening technologies has significantly advanced the understanding of drug resistance mechanisms. For example, genome-wide functional screens in bladder cancer cells identified heterogeneous nuclear ribonucleoprotein U (HNRNPU) as a key regulator of cisplatin resistance. CRISPR-Cas9-mediated HNRNPU ablation significantly restored cisplatin sensitivity, establishing its role in chemoresistance [245]. In parallel, Shalem O et al. employed a CRISPR-Cas9 knockout (GeCKO) library to identify novel mediators of RAF inhibitor resistance in melanoma, unveiling previously unrecognized targets including NF2, CUL3, TADA2B, and TADA1 [246]. Collectively, these findings highlight the potential of CRISPR-based technologies in elucidating resistance mechanisms and informing the development of novel therapeutic strategies.

In the context of ADCs, CRISPR screening has become indispensable for deconvoluting key regulators of drug response. A notable application by Sheng Q et al. leveraged this approach to dissect resistance mechanisms for *PCA062*, a P-cadherin-targeting ADC. Their screen identified MRP1 as a resistance driver while linking SLC46A3 expression to enhanced sensitivity [247], exemplifying how CRISPR can prioritize targets for combinatorial strategies. Furthermore, Lipert et al. performed genome-wide CRISPR/Cas9 screening in HER2-positive BC treated with T-DM1, identifying multiple resistance-associated genes with therapeutic potential [248]. These studies not only deepen our understanding of ADC resistance but also guide rational optimization strategies for next-generation therapeutics.

[Bispecific-targeting ADCs

Tumor heterogeneity presents a substantial hurdle to the efficacy of single-target ADCs. To overcome this, dual or multi-target ADCs have emerged as a promising direction for the next generation of ADCs. These agents, by simultaneously targeting multiple tumor-associated antigens, can significantly enhance the specificity and internalization efficiency of payloads, while may effectively overcoming resistance encountered in current ADCs.

Biparatopic ADCs represent an emerging class of bispecific ADCs that target different epitopes of the same antigen. This design promotes receptor aggregation and enhances target internalization. Currently, there are relatively few biparatopic ADCs in clinical development, with most focusing on HER family targets.

For instance, resistance to HER2-targeted therapies frequently involves compensatory HER3 activation, primarily mediated through heregulin-induced HER2-HER3 heterodimerization and subsequent downstream signaling. Additionally, HER3 upregulation triggers ligand-independent dimerization, further reinforcing tumor survival pathways [249–251]. To address this challenge, bispecific antibodies like *zenocutuzumab* (MCLA-128)-targeting HER2 and HER3-have been developed. This agent not only disrupts HER2-HER3 signaling but also mediates antibody-dependent cellular cytotoxicity (ADCC) [252]. Preclinical study demonstrated that *zenocutuzumab* potently inhibits HER3 and AKT phosphorylation while upregulating pro-apoptotic markers in NRG1 fusion-positive tumors. A pilot clinical study involving chemotherapy-resistant NRG1-altered cancers reported durable responses: two ATP1B1-NRG1-positive pancreatic cancer patients achieved sustained radiologic remission for >12 months; one heavily pretreated CD74-NRG1-positive NSCLC patient achieved PR [253]. Subsequent phase II trial confirmed clinical benefit (mPFS, 6.8 months), with manageable toxicity in advanced NSCLC and pancreatic cancer cohorts [254].

Building on these insights, Zong HF et al. developed a HER2/HER3-targeted biparatopic ADC with potent anti-tumor activity in preclinical settings [255]. Another promising candidate, *REGN5093-M114* (a MET-biparatopic ADC), overcomes MET-driven resistance in EGFR-mutant NSCLC, potentially addressing a key limitation of current MET inhibitors [256]. *IMGN151*, a biparatopic FRα-targeted ADC, highlights this advance. Its optimized design results in longer half-life, higher payload capacity, and improved anti-tumor efficacy over traditional FRα ADCs [257]. A phase I trial is ongoing to evaluate its efficacy in recurrent gynaecological cancers (NCT05527184). Table 3 summarizes the biparatopic ADCs currently in clinical-stage.

Beyond biparatopic formats, bispecific ADCs (bsADCs) targeting distinct antigens have emerged as a promising

Table 3 Ongoing clinical trials evaluating bispecific-target ADCs in cancer

Agent	Target	NCT	Phase	Participants	Trial design	Primary Outcomes
JSKN003	HER2 x HER2	NCT05744427	I/II	Chinese subjects with unresectable locally advanced/metastatic solid tumors	JSKN003	DLT; MTD; ORR
		NCT06846437	III	Unrespectable locally advanced and/or metastatic HER2-positive BC	JSKN003 vs. T-DM1	PFS
		NCT06226766	I/II	Advanced unresectable or metastatic solid malignant tumors (HER2 expression, IHC ≥ 1+)	JSKN003	TEAEs; TRAEs; SAEs; RP2D; DLT; ORR
TQB2102	HER2 x HER2	NCT06452706	II	HER2 negative recurrent/metastatic BC	TQB2102	ORR
		NCT06115902	I	HER2-expressing relapsed/metastatic BC	TQB2102	ORR
		NCT06198751	II	Neoadjuvant treatment of BC with positive HER2 expression	TQB2102	tpCR
		NCT06496490	II	Locally advanced or metastatic NSCLC with HER2 gene abnormality	TQB2102	ORR
		NCT06798207	II	Recurrent/metastatic advanced gynecological tumors	TQB2102	ORR
		NCT06431490	I/II	Recurrent/metastatic advanced HER2-positive BTC	TQB2102	SAEs; RP2D
IMGN151	FRα x FRα	NCT05527184	I	Recurrent gynaecological cancers	IMGN151	SAEs; DLT
BL-B01D1	EGFR x HER3	NCT05983432	I	Metastatic or unresectable NSCLC and other solid tumors	BL-B01D1	DLT; SAEs
		NCT05990803	II	Recurrent or metastatic cervical cancer and other gynecological malignancies	BL-B01D1 vs. SI-B003 vs. BL-B01D1 + SI-B003	ORR; RP2D
		NCT06006169	II	Recurrent or metastatic HNSCC and other solid tumors	BL-B01D1 vs. SI-B003 vs. BL-B01D1 + SI-B003	ORR; RP2D
		NCT05924841	II	Extensive SCLC	BL-B01D1 vs. SI-B003 vs. BL-B01D1 + SI-B003	ORR; RP2D
		NCT06787664	II	Locally advanced or metastatic chordoma	BL-B01D1	ORR
		NCT06042894	II	Unresectable locally advanced or recurrent metastatic HER-2 negative BC	BL-B01D1 vs. SI-B003 vs. BL-B01D1 + SI-B003	ORR; RP2D
		NCT05880706	II	Locally advanced or metastatic NSCLC	BL-B01D1 + Osimertinib	ORR; RP2D
		NCT06598787	II	Recurrent glioblastoma	BL-B01D1	ORR
		NCT06405425	II	Locally advanced or metastatic UC	BL-B01D1 + PD-1	ORR
		NCT06008054	II	Locally advanced or metastatic gastrointestinal tumors	SI-B003 vs. BL-B01D1 + SI-B003 vs. BL-B01D1 + PD-1	ORR; RP2D
		NCT06500026	III	Recurrent SCLC	BL-B01D1 vs. Topotecan	OS
		NCT06471205	II	Advanced or recurrent metastatic TNBC	BL-B01D1 + PD-1	ORR; RP2D
		NCT06382129	III	Advanced EGFR wild-type NSCLC	BL-B01D1 vs. Docetaxel	OS
		NCT06838273	III	EGFR-mutated advanced NSCLC	BL-B01D1 + Osimertinib vs. Osimertinib	PFS
		NCT06382116	III	Advanced NSCLC with EGFR-sensitive mutations	BL-B01D1 vs. Platinum based chemotherapy	PFS
		NCT06382142	III	Advanced or metastatic TNBC	BL-B01D1 vs. Chemotherapy	PFS; OS
		NCT06118333	III	Recurrent or metastatic NPC	BL-B01D1 vs. Chemotherapy	ORR; OS
		NCT06304974	III	Recurrent or metastatic ESCC	BL-B01D1 vs. Chemotherapy	PFS; OS
BL-B16D1	EGFR x HER3	NCT06475131	I	Advanced solid tumors	BL-B16D1	DLT; MTD; RP2D
		NCT06493864	I	Advanced BC and other solid tumors	BL-B16D1	DLT; MTD; RP2D
		NCT06469008	I	Advanced HNSCC and and other solid tumors	BL-B16D1	DLT; MTD; RP2D
AZD9592	EGFR x c-MET	NCT05647122	I	Advanced solid tumors	AZD9592 ± Anti-cancer agents	DLT; SAEs

Table 3 (continued)

Agent	Target	NCT	Phase	Participants	Trial design	Primary Outcomes
ABBV-969	PSMA x STEAP1	NCT06318273	I	mCRPC	ABBV-969	AEs; PSA response rate
JSKN016	Trop2 x HER3	NCT06592417	I	Advanced metastatic solid tumors	JSKN016	TEAEs; TRAEs; SAEs; RP2D; DLT; ORR
IBI3001	EGFR x B7H3	NCT06349408	I	Advanced or metastatic solid tumors	IBI3001	DLT; AEs
IBI3005	EGFR x HER3	NCT06418061	I	Advanced or metastatic solid tumors	IBI3005	DLT; AEs
DM001	EGFR x Trop2	NCT06475937	I	Advanced solid tumors	DM001	DLT; MTD
DB-1419	B7H3 x PD-L1	NCT06554795	I	Advanced solid tumors	DB-1419	SAEs; MTD; RP2D; ORR
DM005	EGFR x c-MET	NCT06515990	I	Advanced solid tumors	DM005	DLT; MTD
ALK202	EGFR x c-MET	NCT06707610	I	Advanced solid tumors	ALK202	MTD
GEN1286	EGFR x c-MET	NCT06685068	I/II	Advanced solid tumors	GEN1286	DLT; AEs
DM002	MUC1 x HER3	NCT06751329	I	Advanced solid tumors	DM001	DLT; MTD
M1231	MUC1 x EGFR	NCT04695847	I	Solid tumors	M1231	DLT; AEs
EBC-129	CEA x CEACAM6	NCT05701527	I	Advanced solid tumors	EBC-129 ± Pembrolizumab	SAEs; MTD; RP2D; ORR

DLT, dose-limiting toxicity; MTD, maximum tolerated dose; ORR, objective response rate; BC, breast cancer; PFS, progression-free survival; TEAEs, treatment-emergent adverse events; TEAEs, treatment-related adverse events; SAEs, serious adverse events; RP2D, recommend phase II dose; tpCR, total physiological complete response; NSCLC, non-small cell lung cancer; BTC, biliary tract cancer; HNSCC, head and neck squamous cell carcinoma; SCLC, extensive stage small cell lung cancer; UC, urothelial carcinoma; OS, overall survival; TNBC, triple-negative breast cancer; PFS, progression-free survival; NPC, nasopharyngeal carcinoma; ESCC, esophageal squamous cell carcinoma; mCRPC, metastatic castration-resistant prostate cancer; PSA, prostate specific antigen

strategy to overcome resistance. For example, HER2 and EGFR are frequently co-expressed, where EGFR amplification or overexpression contributes to resistance against HER2-targeted ADCs [258]. Similarly, amplification of HER2 may lead to resistance to EGFR-targeted therapies [259]. To address these challenges, bsADCs targeting HER2 and EGFR are developed. A case in point is *B2C4-MMAE*, which demonstrated significant anti-tumor activity in models that are unresponsive to either HER2- or EGFR-targeted ADCs. Furthermore, *B2C4-MMAE* also showed remarkable efficacy in patient-derived xenograft (PDX) models that are resistant to T-DXd [260]. *BL-B01D1* is another bsADC targeting EGFR x HER3. It has shown preliminary efficacy despite limited mechanistic characterization [261]. In a phase I trial, *BL-B01D1* elicited an ORR of 34% in heavily pretreated advanced solid tumors (including NSCLC, SCLC, and nasopharyngeal carcinoma), with acceptable safety profile [262].

AZD9592 is also a bsADC that selectively targets EGFR and c-MET. In PDX models of NSCLC and HNSCC, *AZD9592* effectively inhibited tumor growth [263]. Currently, *AZD9592* is under evaluation in a phase I trial for advanced solid malignancies (NCT05647122). Moreover, bsADC targeting PD-L1 and B7-H3 exhibited enhanced durable immunogenic cytotoxicity in laryngeal squamous cell carcinoma models, potentially providing new avenues to overcome immune evasion [264]. Additionally, as previously emphasized, tumor heterogeneity mediates ADCs resistance. To overcome this challenge, Wang M et al. developed an ADC simultaneously targeting MET and RON (PCMbs-MR) for treating cancers with high heterogeneity. Both in vitro and in vivo studies showed that this bsADC has sustained effects on tumors with MET/RON heterogeneity [265]. These advances underscore the expanding landscape of dual-antigen targeted ADCs, as comprehensively summarized in Table 3.

Engineering strategies for ADCs

Current ADC engineering strategies target resistance and efficacy limitations by refining the antibody, linker, and payload. This includes developing novel cytotoxic agents, optimizing antibody specificity, engineering precise delivery systems, and modulating the TME, collectively driving next-generation ADCs with improved potency and selectivity.

Advancements in payload engineering

The mechanisms underlying resistance to conventional ADC payloads have become progressively clearer. Resistance often arises due to mutations in target proteins or the overexpression of drug efflux pumps. To address these challenges, several novel payload strategies have emerged in recent years, some of which hold considerable promise for improving ADCs efficacy.

One promising approach involves targeting RNA, which provides a means of targeting both proliferating and quiescent tumor cells. For instance, amatoxins, such as α-amanitin, selectively inhibit eukaryotic RNA polymerase II, suppressing transcription and protein synthesis. This unique mechanism allows amatoxin-based

Table 4 ADCs combined therapy in various cancer types

Identifier	Phase	Participants	Setting	Trial Design	Outcomes	Ref.
-	-	AML	First-line	GO with intensive chemotherapy	Remission: 86%	[321]
-	-	AML	First relapse	Fractionated doses of GO	CR: 26%	[322]
ISRCTN17161961	-	Young AML patients	Front-line	Adding GO to induction and/or consolidation chemotherapy	GO vs. No GO: CR: 82 vs. 83% OS: 43 vs. 41% RFS: 39 vs. 35%	[323]
-	-	Elderly AML patients	Front-line	Adding GO to induction chemotherapy	GO vs. No GO: ORR: 70 vs. 68%	[325]
-	III	AML	Front-line	Adding low fractionated-dose GO to standard chemotherapy	GO vs. No GO: CR: 812 vs. 75% ($P=0.25$) OS: 53.2 vs. 41.9% ($P=0.0368$) RFS: 50.3 vs. 22.7% ($P=0.0003$)	[326]
NCT01777152	III	PTCL	Front-line	BV + CHP vs. CHOP	5- year: PFS: 51.4 vs. 43.0% OS: 70.1 vs. 61.0%	[329]
NCT01712490	III	Stage III or IV cHL	Front-line	BV + AVD vs. ABVD	5- year PFS: 82.2 vs. 75.3% ($P=0.0017$)	[330]
NCT02166463	III	Pediatric high-risk HL	Front-line	BV + DVPCE vs. ABVE-PC	3- year: EFS: 92.1 vs. 82.5% ($P<0.001$)	[331]
NCT03991884	I	Adults with R/R B-ALL	-	DA-EPOCH + IO	ORR: 83% mOS: 17.0 months mEFS: 9.6 months	[333]
NTR5736	Ib	CD22 + BCP-ALL pediatric patients	-	IO + Vincristine + Dexamethasone + Intrathecal therapy	Pooled response rate: 80%	[334]
NCT03249870	II	Elderly ALL patients	Front-line	IO + Induction chemotherapy	1-year: OS: 73.2%; RFS: 66%; CIR: 25%	[335]
NCT01232556	III	Adults with R/R CD20+/CD2 + aggressive B-NHL	-	R-IO vs. R-B/R-G	mOS: 9.5 vs. 9.5 months ($P=0.708$) mPFS: 3.7 vs. 3.5 months ($P=0.27$)	[336]
NCT01535989	I	R/R CD20 + B-NHL	Subsequent-line	IO + Temsirolimus	PR: 39%	[337]
NCT03274492	III	DLBCL	Front-line	pola-R-CHP vs. R-CHOP	2- year: PFS: 76.7 vs. 70.2% ($P=0.02$) OS: 88.7 vs. 88.6% ($P=0.75$)	[338]
NCT04231877	I	Aggressive large B-cell lymphomas	-	Pola-DA-EPCH-R	Best ORR: 100%; CR: 76%	[339]
NCT04404283	III	DLBCL	-	BV + Len + R vs. Len + R	mOS: 13.8 vs. 8.5 months ($P=0.009$) mPFS: 4.2 vs. 2.6 months ($P<0.001$)	[340]
NCT01896999	I/II	R/R HL	Subsequent-line	Group A: BV + I Group B: BV + N Group C: BV + I + N	A vs. B vs. C: ORR: 76 vs. 89 vs. 82% CR: 57 vs. 61 vs. 73%	[341]
NCT03646123	II	Advanced-stage cHL	-	BV + N + Doxorubicin + Dacarbazine	CR: 88%; ORR: 93%	[342]

Table 4 (continued)

Identifier	Phase	Participants	Setting	Trial Design	Outcomes	Ref.
NCT01716806	II	Advanced-stage cHL	First-line	Part B: BV + DTIC Part D: BV + N	CR: Part B: 64%; Part D: 67% mPFS: Part B: 47.2 months; Part D: NA	[343]
NCT04484623	III	Multiple myeloma	Subsequent-line	BPd vs. PVd	12- month PFS: 71 vs. 51% ($P<0.001$)	[344]
NCT03848845	I/II	Multiple myeloma	Subsequent-line	Belantamab mafodotin + Pembrolizumab	ORR: 47%	[345]
NCT00893399	III	NPM1-mutated AML	First-line	Induction therapy plus ATRA followed by consolidation therapy, without or with GO	GO vs. No GO: Short-term EFS: 58 vs. 53% ($P=0.10$) 2- year CIR: 25 vs. 37% ($P=0.0028$)	[346]
NCT00951665	Ib/IIa	HER2 + advanced BC	-	T-DM1 + Paclitaxel ± Pertuzumab	ORR: 50.0% CBR: 56.8%	[347]
NCT02131064	III	HER2 + BC	Neoadjuvant	T- DM1/P vs. Trastuzumab + Pertuzumab + Chemotherapy	pCR: 44.4 vs. 55.7% ($P=0.016$)	[350]
NCT01042379	II	Stage II/III HER2 + BC	Neoadjuvant	T-DM1/P vs. THP vs. TH	pCR: 63 vs. 72 vs. 33%	[351]
NCT01120184	III	HER2 + advanced BC	First-line	HT vs. T-DM1 vs. T-DM1/P	PFS: 13.7 vs. 14.1 vs. 15.2 months	[352]
NCT01983501	Ib	ERBB2/HER2 + BC	Subsequent-line	Tucatinib + T-DM1	mPFS: 8.2 months; ORR: 47%	[354]
NCT02073916	Ib	HER2 + advanced BC	-	T-DM1 + Lapatinib + Nab-paclitaxel	ORR: 85.7%	[355]
NCT04039230	Ib	mTNBC	-	sequential SG/TZP	PFS: 7.6 months	[357]
-	I	Trastuzumab- and taxane-resistant HER2 + MBC	Subsequent-line	T-DM1 + Alpelisib	ORR: 43%; mPFS: 8.1 months	[358]
NCT02073487	II	Early-stage HER2 + BC	Neoadjuvant	T-DM1 + Lapatinib + Nab-paclitaxel vs. THP	RCB 0–1: 100 vs. 62.5% ($P=0.0035$)	[359]
NCT02236000	Ib/II	HER2 + MBC	Subsequent-line	T-DM1 + Neratinib	ORR: phase I: 63% phase II: 32%	[360]
NCT01494662	II	HER2 + BCBMs	-	T-DM1 + Neratinib: cohort 4 A: previously untreated BCBM cohort 4B: progressing after local CNS-directed therapy without T-DM1 cohorts 4 C: progressing after local CNS-directed therapy with prior exposure to T-DM1	CNS ORR: 4 A vs. 4B vs. 4 C: 33.3 vs. 35.3 vs. 28.6%	[361]
NCT01702558	II	MBC	Subsequent-line	T-DM1 + Capecitabine vs. T-DM1	ORR: 44 vs. 36% ($P=0.34$)	[362]
NCT02924883	II	HER2 + advanced BC	Subsequent-line	T-DM1 + Atezolizumab vs. T-DM1	mPFS: 8.2 vs. 6.8 months ($P=0.33$)	[363]
NCT03032107	Ib	HER2 + MBC	Subsequent-line	T-DM1 + Pembrolizumab	ORR: 20%; mPFS: 9.6 months	[364]
NCT03523572	Ib	HER2 + MBC	Subsequent-line	T-DXd + Nivolumab Cohort 1: HER2-positive Cohort 2: HER2-low	CBR: Cohort 1: 71.9% Cohort 2: 56.3%	[365]
NCT05629585	III	TNBC	Adjuvant	Dato-DXd + Durvalumab vs. Dato-DXd vs. ICT	Unrevealed	[367]

Table 4 (continued)

Identifier	Phase	Participants	Setting	Trial Design	Outcomes	Ref.
NCT06103864	III	PD-L1-high locally recurrent inoperable or metastatic TNBC	-	Dato-DXd + Durvalumab vs. ICT + Pembrolizumab vs. Dato-DXd	Unrevealed	[368]
NCT02996825	I	FRα + EOC, EC, or TNBC	Subsequent-line	MIRV + GEM	achieved PR: 45% confirmed PR: 15%	[369]
NCT02751918	Ib	Platinum-resistant EOC	Subsequent-line	Anetumab ravtansine + PLD	ORR: 27.7%; mPFS: 8.5 months	[370]
NCT02606305	Ib	FRα + platinum-resistant OC	Subsequent-line	MIRV + Bevacizumab	ORR: 39%; mPFS: 6.9 months	[371]
NCT02606305	Ib	Recurrent, platinum-sensitive OC	Subsequent-line	MIRV + Carboplatin + Bevacizumab	ORR: 83%; mPFS: 13.5 months	[372]
NCT03786081	Ib/II	r/mCC	-	arm D: TV + Carboplatin (1 L) arm E: TV + Pembrolizumab (1 L) arm F: TV + Pembrolizumab (2 L/3L)	ORR: D vs. E vs. F: 54.5 vs. 40.6 vs. 35.3%	[373]
NCT02099058	Ib	c-Met + NSCLC	Subsequent-line	Teliso-V + Erlotinib	mPFS: 5.9 months	[374]
NCT02099058	Ib	Advanced EGFR-mutated, c-Met+, non-squamous NSCLC	Subsequent-line	Teliso-V + Osimertinib	ORR: 50%; mPFS: 7.4 months	[375]
-	Ib/II	LA/mUC	First-line	EV + Pembrolizumab	ORR: 73.3%; mOS: 26.1 months	[376]
NCT03547973	II	mUC	Subsequent-line	SG + Pembrolizumab	ORR: 41%; mPFS: 5.3 months	[377]
NCT04526691	Ib	NSCLC	Subsequent-line	Dato-DXd + Pembrolizumab ± Pt-CT	ORR: Doublet: 38%; Triplet: 47% mPFS: Doublet: 10.8 months Triplet: 7.8 months	[378]
NCT05215340	III	PD-L1-high advanced NSCLC without AGA	First-line	Dato-DXd + Pembrolizumab vs. Pembrolizumab	Unrevealed	[379]
NCT05555732	III	PD-L1-low advanced NSCLC without AGA	First-line	Dato-DXd + Pembrolizumab ± Chemotherapy vs. Pembrolizumab + Chemotherapy	Unrevealed	[380]
-	II	prrHGOC	Subsequent-line	ARB vs. PB	mPFS: 5.3 vs. 12.7 months ($P = 0.03$) ORR: 21 vs. 65%	[383]
NCT03992131	Ib	Cancers with or without mutations in HRR genes	-	SG + Rucaparib	PR: 50%	[384]
-	Ib/II	LA/mUC	-	DV + Toripalimab	ORR: 73.2%; mPFS: 9.3 months mOS: 33.1 months	[385]
NCT03126630	I/II	Mesothelin + pleural mesothelioma	Subsequent-line	AR + Pembrolizumab vs. Pembrolizumab	mPFS: 12.2 vs. 3.9 months ($P = 0.20$)	[386]
NCT02341625	I/IIa	Advanced solid tumors	-	BMS-986,148 monotherapy vs. BMS-986,148 + Nivolumab	Unrevealed	[387]
NCT06667167	II	ES-SCLC	First-line	Induction therapy consisting of pembrolizumab + carboplatin + etoposide, then followed by maintenance therapy combining pembrolizumab + SG	Unrevealed	[388]
NCT04879329	II	HER2 + LA/mUC	-	DV vs. DV ± Pembrolizumab	Unrevealed	[389]

Table 4 (continued)

Identifier	Phase	Participants	Setting	Trial Design	Outcomes	Ref.
NCT04863885	I/II	mUC	First-line	SG + I + N	ORR: 66.6%; mPFS: 8.8 months	[390]
NCT05186974	II	advanced NSCLC	First-line	SG + Pembrolizumab ± Carboplatin/cisplatin	Unrevealed	[391]
NCT04612751	Ib	advanced NSCLC	-	Dato-DXd + Immunotherapy ± Carboplatin	Unrevealed	[392]
NCT03924895	III	MIBC	Perioperative	Pembrolizumab vs. Pembrolizumab + EV vs. RC + PLND alone	Unrevealed	[393]
NCT04960709	III	MIBC	Neoadjuvant	Arm 1: D (anti–PD-L1) + T (anti–CTLA-4) + EV Arm 2: D + EV Arm 3: no neoadjuvant treatment	Unrevealed	[394]
NCT04700124	III	MIBC	Perioperative	Neoadjuvant EV + Pembrolizumab followed adjuvant EV + Pembrolizumab after RC + PLND vs. Neoadjuvant cisplatin-based chemotherapy followed by observation after RC + PLND	Unrevealed	[395]
NCT03288545	Ib/II	LA/mUC	First-line	EV + Pembrolizumab vs. EV	ORR: 64.5 vs. 45.2%	[396]

GO, gemtuzumab ozogamicin; AML, acute myeloid leukemia; mRFS, median relapse-free survival; CR, complete remission/response; OS, overall survival; ORR, overall response rate; PTCL, peripheral T-cell lymphoma; BV, brentuximab vedotin; CHP, cyclophosphamide, doxorubicin, and prednisone; CHOP, cyclophosphamide, doxorubicin, vincristine, and prednisone; PFS, progression-free survival; HL, hodgkin lymphoma; AVD, doxorubicin, vinblastine, and dacarbazine; ABVD, doxorubicin, bleomycin, vinblastine, and dacarbazine; DVPCE, doxorubicin, vincristine, etoposide, prednisone, and cyclophosphamide; ABVE-PC, doxorubicin, bleomycin, vincristine, etoposide, prednisone, and cyclophosphamide; EFS, event-free survival; B-ALL, B-cell acute lymphoblastic leukemia or lymphoma; IO, inotuzumab ozogamicin; DA-EPOCH, dose-adjusted etoposide, prednisone, vincristine, cyclophosphamide, and doxorubicin; BCP-ALL, B-cell precursor acute lymphoblastic leukemia; CIR, cumulative incidence of relapse; R/R B-NHL, relapsed/refractory B-cell non-Hodgkin lymphoma; R-IO, rituximab, inotuzumab ozogamicin; R-B, rituximab, bendamustine; R-G, rituximab, gemcitabine; PR, partial remission/response; DLBCL, diffuse large B-cell lymphoma; pola-R-CHP, polatuzumab vedotin, rituximab, cyclophosphamide, doxorubicin, and prednisone; R-CHOP, rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone; Pola-DA-EPCH-R, polatuzumab vedotin, etoposide, prednisone, cyclophosphamide, doxorubicin, and rituximab; BV + Len + R, brentuximab vedotin, lenalidomide, rituximab; I, ipilimumab; N, nivolumab; cHL, classical Hodgkin lymphoma; DTIC, Dacarbazine; NA, not reached; BPd, belantamab mafodotin, pomalidomide, and dexamethasone; PVd, pomalidomide, bortezomib, and dexamethasone; Belamaf, belantamab mafodotin; ATRA, all-trans retinoic acid; BC, breast cancer; CBR, clinical benefit rate; MBC, metastatic breast cancer; NPLD, non-pegylated liposomal doxorubicin; DLT, dose-limiting toxicity; pCR, pathological complete response; T-DM1/P, T-DM1, pertuzumab; THP, paclitaxel, trastuzumab and pertuzumab; TH, paclitaxel/trastuzumab; HT, trastuzumab, taxane; SG, sacituzumab govitecan; TZP, talazoparib; mTNBC, metastatic triple-negative breast cancer; RCB, residual cancer burden; BCBMs, breast cancer brain metastases; CNS, central nervous system; PCbHP, nab-paclitaxel combined with carboplatin, trastuzumab, and pertuzumab; ICT, investigator's choice of therapy; MIRV, mirvetuximab soravtansine; GEM, gemcitabine; EOC, epithelial ovarian cancer; EC, endometrial cancer; PLD, pegylated liposomal doxorubicin; OC, ovarian cancer; r/mCC, recurrent or metastatic cervical cancer; TV, tisotumab vedotin; 1 L, first-line; 2 L/3L, second-/third-line; NSCLC, non-small-cell lung cancer; Teliso-V, telisotuzumab vedotin; LA/mUC, locally advanced or metastatic urothelial cancer; EV, enfortumab vedotin; Pt-CT, cisplatin 75 mg/m2 or carboplatin AUC 5; AGA, actionable genomic alterations; HGOC, high-grade serous/endometrioid ovarian cancers; prrHGOC, platinum-resistant/refractory HGOC; AR, Anetumab ravtansine; ARB, AR, bevacizumab; PB, paclitaxel, bevacizumab; HRR, homologous recombination repair; DV, disitamab vedotin; ES-SCLC, extensive-stage small cell lung cancer; MIBC, muscle-invasive bladder cancer; RC + PLND, radical cystectomy + pelvic lymph node dissection

ADCs to demonstrate cytotoxic effects not only against actively dividing tumor cells but also against dormant populations, including cancer stem cells. Importantly, RNA polymerase II plays a central role in cellular metabolism, significantly reducing tumor cells' ability to bypass this pathway. Moreover, α-amanitin resists efflux by classical drug resistance mechanisms-most notably the MDR1 pump, an ATP-driven efflux transporter that actively expels many conventional chemotherapeutics. This unique property enables α-amanitin-based ADCs to overcome a major resistance pathway that limits traditional ADCs [266].

The efficacy of amanitin-conjugated ADCs has been demonstrated across preclinical models, including in vitro and in vivo. For instance, *chiHEA125-Ama* (a chimeric anti-EpCAM antibody conjugated to α-amanitin) potently inhibited proliferation in diverse tumor cells. Notably, in pancreatic cancer xenograft models, this ADC induced complete tumor regression in 90% of treated mice [267]. Similarly, *HDP-101*, a BCMA-targeting antibody-drug conjugate armed with amanitin, has shown promising therapeutic potential, demonstrating robust preclinical activity and early clinical efficacy in multiple myeloma. Preclinical studies revealed its dual ability to eradicate both proliferating and quiescent myeloma cells with high selectivity, while subcutaneous and disseminated xenograft models showed that low doses induced durable complete remissions [268]. Interim results from the ongoing phase I trial (NCT04879043) demonstrate dose-dependent activity in relapsed/refractory patients. At the 100 μg/kg dose level, investigators observed an ORR of 50%, which included one CR. The safety profile was characterized by manageable hematologic toxicities, primarily grade 3/4 thrombocytopenia with full reversibility, and no treatment-related discontinuations at this dose [269]. *HDP-101* also displayed enhanced efficacy in molecularly defined subtypes, particularly in overcoming adhesion-mediated drug resistance and excelling in high-risk del17p myeloma [270]. Additionally, TROP2-targeted amanitin ADCs (ATAC) have shown superior anti-tumor activity compared to conventional ADCs like SG in xenograft models of refractory pancreatic cancer and TNBC, warranting further investigation in TROP2-expressing malignancies [271].

Another promising class of payloads involves RNA splicing inhibitors, which exploit tumor addiction to splicing machinery by inducing lethal intron retention and exon skipping in oncogenic transcripts (e.g., MYC, BCL2), thereby achieving broader therapeutic windows than conventional chemotherapy [272]. *Thailanstatins*, potent inhibitors of the mRNA spliceosome, have been conjugated to antibodies to enhance therapeutic efficacy [273]. In a study performed by Puthenveetil S et al., a *thailanstatin* derivative was conjugated to *trastuzumab*, generating series of *trastuzumab-thailanstatin* ADCs that demonstrated superior activity compared to T-DM1 in HER2-high GC models, including multidrug-resistant lines. In N87 xenograft models, dose-response analyses established *thailanstatin*-conjugated ADCs (3 mg/kg) as pharmacodynamically superior to T-DM1 [274]. However, further validation is needed before these findings can be translated into clinical practice.

Another innovative class of payloads targets B-cell lymphoma-extra large (Bcl-xL) inhibitors, which counteract apoptosis resistance in both solid and hematologic malignancies. For instance, *mirzotamab cletuzoclax* (ABBV-155), an ADC targeting B7-H3 (CD276) conjugated to a Bcl-xL inhibitor, has shown promising anti-tumor activity in early-phase trials for the treatment of relapsed and refractory solid tumors (NCT03595059) [275]. Preclinical studies further validated its potential in glioblastoma [276], highlighting the growing interest in combining Bcl-xL inhibition with ADC-based therapies to overcome resistance mechanisms by inducing synergistic apoptosis.

Dual-payload ADCs, which incorporate two distinct cytotoxic agents on a single antibody, represent a frontier in ADC design aimed at enhancing efficacy and overcoming resistance. For instance, Yamazaki CM et al. developed an anti-HER2 ADC co-conjugated with MMAE and MMAF, demonstrating synergistic tumor suppression in HER2-heterogeneous BC models through enhanced bystander effects [277]. Similarly, a site-specifically conjugated HER3-targeting ADC bearing both a topoisomerase I inhibitor and an EGFR-TKI showed promising tumor regression in EGFR-TKI-resistant NSCLC PDX models without significant toxicity [278]. However, not all dual-payload combinations exhibit synergy. For example, combinations of MMAF/PNU-159,682 and MMAE/PBD did not display cooperative cytotoxicity in anti-HER2 ADCs [279, 280]. While these constructs show potential, challenges remain regarding manufacturability, safety, and optimizing payload combinations. Despite these advances, critical challenges persist-particularly in manufacturability, safety, and payload optimization. Specifically, dual-payload ADCs demand far more complex bioconjugation processes than single-payload formats, introducing scalability hurdles during production. Furthermore, increasing the overall drug-to-antibody ratio (DAR) often exacerbates hydrophobicity and aggregation risks, which can ultimately compromise pharmacokinetic profiles and clinical tolerability [281].

Efforts has also shifted toward molecular-level optimization strategies, particularly through structural modification of payloads. A key study by Johansson MP et al. employed nuclear magnetic resonance (NMR) spectroscopy to analyze MMAE and MMAF conformations, revealing that nearly half of these payloads adopt inactive states in solution, which reduces ADC efficacy and contributes to off-target toxicity. Strategic modifications (e.g., aromatic ring substitution, (3)/(4)-amino-group alteration, or dolaproine-dolaisoleuine amide bond optimization) can favor bioactive conformations, lower isomer interconversion barriers, and ultimately enhance potency. Although further clinical validation is required, this rational payload design not only represents a viable strategy to overcome resistance mechanisms, but also provides a framework for next-generation ADCs with improved therapeutic windows, reduced systemic toxicity, and broader clinical applicability. Specifically, such fine-tuning of payload chemistry may lead to (1) enhanced tumor selectivity by stabilizing drug-target interactions, (2) decreased off-target effects through controlled payload release kinetics, and (3) expanded treatment opportunities for resistant cancers previously deemed unsuitable for ADC therapy [282].

Furthermore, immune-stimulating antibody conjugates (ISACs) represent an innovative class of ADCs that combine the targeting precision of antibodies with immune system-modulating small molecules. These ADCs leverage the innate and adaptive immune responses to enhance tumor destruction. Several new immune-modulating payloads are under investigation, including toll-like receptor (TLR) agonists and stimulator of interferon genes (STING) agonists, both of which are designed to enhance antigen presentation and T-cell activation within the TME [283]. For example, *SBT6050*, an anti-HER2 ADC conjugated to a TLR8 agonist, demonstrated HER2-dependent immunotherapeutic effects in vivo [284]. Similarly, *HE-S2*, an anti-PD-L1 ADC conjugated to a TLR7 agonist, not only blocks PD-1/PD-L1 interactions but also activates TLR7/8 signaling, inducing the upregulation of PD-L1 expression via IFN-γ, thereby sensitizing tumors to immune checkpoint blockade [285]. Additionally, STING agonists have shown great promise in enhancing immune responses. *XMT-2056*, a HER2-targeted ADC conjugated to a STING agonist, induced ICD in both HER2-positive and bystander HER2-negative cells, with preclinical data supporting a synergistic effect when combined with anti-PD-1 therapy [286]. These studies suggest a novel approach to enhance ADC-mediated immune responses and suggest a potential paradigm shift in cancer immunotherapy.

Overcoming drug delivery barriers

Drug delivery barriers represent significant challenges in ADCs therapy, including mechanisms such as efflux pump-mediated drug clearance and toxin inactivation due to lysosomal sequestration. These barriers limit the efficacy of ADCs and contribute to the development of drug resistance. Recent advances in precision inhibition of efflux pumps and optimization of lysosomal escape have significantly improved ADCs delivery efficiency, thereby mitigating some of the resistance mechanisms that hinder therapeutic outcomes.

To address the inefficient intracellular processing of ADCs, researchers proposed strategies to improve internalization, lysosomal targeting, and intratumoral processing. For instance, de Goeij BE et al. developed a bispecific antibody (bsAb)-based ADC where one of the bsAb arms specifically targets CD63, a protein capable of shuttling between the plasma membrane and intracellular compartments to promote HER2-specific internalization and lysosomal delivery. In vitro studies showed that bsHER2 x CD63 could be effectively transported into the lysosomes of HER2-positive tumor cells. Subsequently, bsHER2 x CD63 was conjugated with the microtubule disruptor duostatin-3 to create the bsHER2xCD63-ADC, which demonstrated potent cytotoxicity in HER2-positive cell lines, whereas ADC targeting HER2 or CD63 alone had no impact on tumor growth. These findings validate CD63 targeting as a strategy to enhance payload delivery in poorly internalizing ADCs [287].

Similarly, Sortilin-1 (SORT1), a transmembrane protein involved in regulating protein transport through endocytosis, has been explored as a target to improve ADC internalization. Zhuang W et al. developed a HER2/SORT1-targeting bispecific ADC, which demonstrated strong binding and internalization in HER2-low tumor cells. Upon conjugation with DXd, this bispecific antibody, termed *bsSORT1×HER2-DXd*, exhibited significant cytotoxicity against HER2-low tumors and potent anti-tumor efficacy in vivo [288]. In another innovative approach, the authors engineered a bispecific anti-SLC3A2/PD-L1 ADC (*SLC3A2/PD-L1* bsADC-MMAE). This conjugate not only blocked the PD-1/PD-L1 interaction but also enhanced the lysosomal degradation of internalization-refractory PD-L1 antibodies. Compared to monospecific ADCs, the *SLC3A2/PD-L1* bsADC showed superior efficacy in PD-L1-low tumors. These findings highlight the potential of lysosome-targeting bispecific antibodies to enhance ADCs potency and overcome therapeutic resistance [289].

Another critical limitation of conventional ADCs lies in their reliance on the endosomal–lysosomal pathway for intracellular toxin release. Although this route facilitates internalization, it also introduces multiple inefficiencies. First, premature cleavage of the linker within the lysosome may lead to unintended release of the cytotoxic agent, thereby reducing the specificity of drug delivery. Second, enzymatic degradation of the payload in the lysosomal environment can significantly diminish its functional activity [290–292]. These drawbacks have prompted the exploration of alternative delivery strategies aimed at improving the precision, stability, and on-target activation of payloads. One such innovative approach is fluorescence imaging-guided drug delivery, which leverages high spatiotemporal resolution for real-time visualization of therapeutic distribution. For example, Del Valle et al. designed a gold nanostar-based delivery system (dsDDA-AuNS) conjugated with the anti-nucleolin aptamer AS1411. *Doxorubicin* was site-specifically tethered to deoxyguanosine residues via methylene linkers. Upon near-infrared (NIR) laser irradiation, dsDDA-AuNS facilitated controlled drug release in chemoresistant BC models, producing a synergistic photothermal and chemotherapeutic effect. This dual-modality strategy demonstrated superior tumor suppression compared to monotherapy alone [293].

In parallel, efforts have been made to incorporate real-time drug tracking into ADCs through fluorescent probes. Thankarajan E et al. reported the development of an antibody-drug conjugate integrated with a drug-release-sensitive NIR dye (IRD-Ab-mXCy-CLB). This system enabled dynamic imaging of drug accumulation in HER2-positive BC models. Notably, the ADCs showed accelerated tumor targeting with minimal off-target release, underscoring the advantage of incorporating diagnostic functionalities into therapeutic platforms [294].

Despite these promising outcomes, the clinical translation of light-responsive systems is potentially limited by tissue penetration depth and dependence on external activation sources. Consequently, emerging pH- and reactive oxygen species (ROS)-responsive delivery platforms are garnering interest due to their capacity for tumor-selective activation. Particularly, nanocarriers designed with pH/ROS or pH/light dual-responsive features have demonstrated enhanced drug release under tumor-specific microenvironmental conditions. These systems not only improve delivery precision but also potentiate drug activity by minimizing systemic exposure and maximizing local bioavailability [295–298]. Such intelligent delivery platforms offer promising potential to bypass the fundamental challenges confronting conventional ADCs and photodependent nanomedicines.

TME reprogramming

Although the resolution of intracellular delivery barriers has significantly advanced the development of ADCs, overcoming the substantial external challenge posed by the TME remains a critical hurdle. The

immune-suppressive properties and physical barriers, such as dense ECM networks, severely limit the permeability of therapeutic agents, thereby reducing the overall efficacy of ADCs. To address these challenges, researchers are adopting integrated strategies that target both the cellular and extracellular components of the TME.

One of the major obstacles to efficient drug delivery is the ECM, which, due to its dense collagen network and stromal hypertension, significantly impedes drug penetration. This limitation is particularly relevant in the context of various malignancies, including pancreatic cancer, NSCLC, and BC. Fibronectin extra domain B splice variant (EDB + FN), a protein secreted by CAFs, is highly expressed in the stroma of these cancers and plays a pivotal role in regulating tumor growth, angiogenesis, and invasion [299, 300]. In response to this discovery, researchers developed an EDB + FN-targeted ADC (*EDB-ADC*). Preclinical study has demonstrated that *EDB-ADC* accumulates precisely within the TME and induces mitotic arrest in tumor cells. In murine models of lung and pancreatic cancer, *EDB-ADC* effectively suppressed tumor growth, likely due to enhanced intracellular payload penetration and bystander effects. Notably, *EDB-ADC* also upregulated PD-L1 expression and promoted the infiltration of CD3+ T cells, providing a rationale for combining it with PD-L1 inhibitors. In subsequent investigations in BC model, the combination of EDB-ADC with anti-PD-L1 therapy induced durable anti-tumor immunity, highlighting the potential for synergistic immunotherapeutic strategies [300].

In the context of stroma-rich PDAC, Tsujii S et al. developed a glypican-1 (GPC1)-targeted ADC (*GPC1-ADC*). GPC1 is specifically expressed on PDAC-derived CAFs, making it an ideal target for delivering cytotoxic payloads. By targeting both GPC1+ CAFs and tumor cells, *GPC1-ADC* exhibited potent anti-tumor activity both in vitro and in vivo. Further studies revealed that *GPC1-ADC* is internalized by CAFs, where the released MMAE payload, mediated by the MDR1, induces apoptosis in neighboring tumor cells. These findings demonstrate the therapeutic potential of TME-targeted ADCs through their ability to disrupt dense stromal barriers, reprogram the immunosuppressive TME, and significantly improve payload delivery to previously inaccessible tumor cells [301]. Such multimodal targeting enables deeper intratumoral drug distribution, resulting in enhanced cytotoxic efficacy, particularly valuable for treating stroma-rich tumors that typically exhibit treatment resistance.

Another promising strategy to improve ADC delivery efficiency involves vascular normalization, which has emerged as a potential breakthrough. This approach targets vascular endothelial growth factor (VEGF) signaling, a key driver of tumor vascular abnormalities, by simultaneously suppressing pathological angiogenesis, reducing vascular permeability, and inhibiting metastasis [302]. Inhibition of VEGF using monoclonal antibodies, such as *bevacizumab*, promotes vascular normalization through several mechanisms, including the pruning of immature tumor vasculature and the reinforcement of remaining vessels [303, 304]. Building on this, Li Y et al. engineered a *bevacizumab*-based ADC (*Vedotin*) to overcome the limitations of traditional *bevacizumab* combined with chemotherapy. Preliminary data revealed that *Vedotin* exhibited excellent stability, rapid payload release, and simultaneous anti-tumor and antiangiogenic effects, thus representing a promising avenue for improving ADC therapeutic outcomes [305].

It is important to note that the increased stiffness of the ECM within the TME poses a significant barrier to ADC delivery. This stiffening is primarily driven by aberrant activation of the Rho-associated protein kinase (ROCK) pathway. As a downstream effector of Rho GTPase, ROCK enhances the contractility of actin filaments through phosphorylation of myosin light chain (MLC) and LIM kinase (LIMK). This process not only augments the contractility of tumor cells but also promotes collagen fiber remodeling and crosslinking, ultimately resulting in ECM stiffening [306–308]. In this context, ROCK inhibitors such as *fasudil* have shown promise in reversing drug resistance and sensitizing tumor cells to chemotherapy [309–311]. Notably, targeting ROCK2 has been demonstrated to normalize ECM tissue, significantly improving nanoparticle penetration in pancreatic cancer [312]. While the evidence supporting ROCK inhibitors in overcoming ADCs resistance remains limited, these findings provide new insights into potential strategies for optimizing ADCs delivery.

In addition to physical barriers, the immune-suppressive characteristics of the TME further attenuate ADC efficacy. To counteract this limitation, researchers have turned to ISACs. As the next generation of ADCs, ISACs combine dual therapeutic functions: they activate innate immune responses while also stimulating adaptive immune responses. A study by Ackerman SE et al. on an HER2-targeted ISAC demonstrated that localized immune activation within the TME could drive tumor eradication and establish immunological memory. Remarkably, even in murine models with low HER2 expression and *trastuzumab* resistance, ISAC significantly suppressed tumor growth [313]. Subsequent preclinical studies have further validated the feasibility of ISACs as a promising therapeutic strategy [314–316].

Despite their potential, the clinical application of ISACs faces several challenges. For instance, *NJH395*, an ISAC combining HER2 antibodies with TLR7 agonists, was evaluated in a phase I trial (NCT03696771) for the treatment of non-breast HER2-positive cancers. Although it

could induce Type I interferon responses, the efficacy was limited, with no observed response (ORR = 0%). Moreover, the rapid and robust systemic immune activation led to cytokine release syndrome (CRS) in 55.6% of patients [317]. While ISAC development remains in its early stages, optimizing agonist selection and dosing strategies may unlock their full therapeutic potential.

Finally, antigen presentation by myeloid cells within the TME is often suppressed, which poses an additional challenge to effective immunotherapy. The CD47-SIRPα signaling pathway plays a crucial role in inhibiting macrophage activation by providing a 'don't eat me' signal. Blocking this pathway has been shown to significantly enhance macrophage-mediated phagocytosis of tumor cells, thereby promoting immune-mediated anti-tumor effects [318]. Based on this, Si Y et al. developed an ADC targeting CD47, which demonstrated robust tumor suppression and well tolerance in a chemotherapy-pretreated TNBC murine model [319]. Another promising approach involves combining SIRPα antibodies, such as *DS-1103a*, with ADCs. Studies have shown that this combination significantly enhances antibody-dependent cellular phagocytosis by inhibiting the CD47-SIRPα pathway, resulting in superior anti-tumor activity in xenograft models compared to single-agent therapies [320]. Together, these studies provide strong evidence supporting the use of ADC-based CD47-SIRPα blockade strategies for improving therapeutic outcomes in cancer.

In summary, overcoming the barriers posed by the TME remains critical for maximizing the efficacy of ADCs. Continued integration of TME-modulating strategies with ADC technologies holds promise for enhancing drug delivery and overcoming resistance. These advances may ultimately broaden the clinical applicability of ADCs across a wider spectrum of solid cancers.

Exploration of ADCs combination therapies

Advancements in understanding resistance mechanisms have catalyzed the development of innovative strategies combining ADCs with chemotherapy, targeted agents, and immunotherapies. These combinations substantially expand the therapeutic potential of ADCs by overcoming the efficacy limitations of conventional treatments. Increasing evidence supports that coordinated multi-pathway inhibition can break through efficacy plateaus, offering enhanced clinical outcomes. Although phase II/III trials have demonstrated survival benefits across a variety of malignancies, the clinical translation of these therapies varies considerably depending on the combinatorial synergies involved. This section critically discusses breakthrough advances and persisting challenges in hematologic and solid tumors.

Hematologic malignancies

The clinical development of GO, the first FDA-approved ADC, exemplifies the evolution of precision oncology, particularly in AML. Initial discordances observed across several randomized trials were resolved by a meta-analysis conducted by Hills et al. [321–327], which established that the incorporation of GO into induction chemotherapy significantly reduces the risk of relapse (P = 0.0001) and improves 5-year OS (P = 0.01), especially in patients with favorable cytogenetic profiles [328]. These findings not only reconciled prior controversies but also initiated a paradigm shift towards molecularly stratified therapies for AML.

The landscape of lymphoma treatment has also been significantly altered by the use of BV. The ECHELON-2 trial demonstrated that BV combined with *cyclophosphamide*, *doxorubicin*, and *prednisone* (A + CHP) significantly improved mPFS compared to CHOP (62.3 vs. 23.8 months, P = 0.0077) after 5 years of follow-up, with no increase in treatment-related mortality [329]. In HL, BV-chemotherapy combination has shown significant improvements in 5-year PFS rate (82.2% vs. 75.3%, P = 0.0017) [330]. Cross-population validation further highlighted that incorporating BV into chemotherapy regimens resulted in a remarkable 59% reduction in mortality risk among pediatric HL patients [331], while maintaining robust efficacy and tolerability in older cohorts [332]. Moreover, CD22-targeted ADC has emerged as a breakthrough in the treatment of ALL, with the latest clinical data demonstrating consistent efficacy across adult, pediatric, and elderly populations, alongside manageable toxicity profiles [333–335]. Collectively, these findings underscore the transformative potential of ADCs in precision oncology, utilizing target-selective payload delivery to achieve significant clinical benefits while minimizing the systemic toxicity often associated with conventional chemotherapy.

However, the clinical efficacy of combining ADCs with targeted agents has demonstrated notable variability across different cancer types and regimens. For instance, a phase III trial conducted by Dang NH et al. showed that the combination of IO with *rituximab* (R-InO) did not significantly improve OS (HR, 1.04, P = 0.708) or PFS (HR, 1.18, P = 0.27) compared to *rituximab-gemcitabine* (R-G) in relapsed/refractory aggressive B-cell non-Hodgkin lymphoma (B-NHL) [336]. Similarly, while the combination of IO with *temsirolimus* induced partial responses in 39% of CD22-positive refractory B-NHL patients, its clinical application was limited by dose-limiting toxicities [337]. The observed toxicity may, in part, stem from insufficient CD22-mediated internalization, potentially leading to extracellular payload release and off-target effects. However, other mechanisms could contribute to these adverse events, and further investigation

is required to identify the underlying causes of treatment-related toxicity.

In contrast, *polatuzumab vedotin* (PV) has demonstrated significant clinical efficacy in the treatment of DLBCL. The pivotal POLARIX trial showed that PV combined with *rituximab*, *cyclophosphamide*, *doxorubicin*, and *prednisone* (pola-R-CHP) improved PFS rates by 9.5% (76.7% vs. 70.2%, $P = 0.02$) and reduced the risk of disease progression or relapse by 27% compared to standard R-CHOP chemotherapy [338]. Building on this success, an intensified PV-based regimen (Pola-DA-EPCH-R) was evaluated in aggressive large B-cell lymphomas, yielding promising initial results with a 100% ORR and a 76% CR rate among 17 evaluable patients [339]. Similarly, the combination of BV, *rituximab*, and *lenalidomide* (BV + Len + R) demonstrated survival benefits and manageable toxicity in relapsed or refractory DLBCL [340]. Although ADC-chemotherapy combination regimens present significant toxicity management challenges, strategically optimized dosing approaches have shown synergistic benefits in clinical trials.

Simultaneously, the immunomodulatory potential of ADCs has led to the emergence of a novel therapeutic paradigm through their combination with immunotherapy. In hematologic malignancies, the landmark E4412 trial demonstrated that the combination of BV and *nivolumab* (B + N) achieved an ORR of 89% with only 16% grade 3–4 adverse events (AEs) in relapsed/refractory HL. In contrast, the triple combination with *ipilimumab* (B + N + I) yielded an ORR of 82%, albeit with a higher toxicity profile (50% of patients experiencing grade ≥ 3 AEs) [341]. A subsequent phase II trial (NCT01896999) is currently comparing the efficacy and safety profiles of these two leading regimens (B + N vs. B + N + I). Additionally, a separate phase II study validated the efficacy of the B + N combination with *doxorubicin* and *dacarbazine* as frontline therapy for advanced classical HL, achieving striking complete and overall response rates of 88% and 93%, respectively. Grade ≥ 3 AEs were predominantly limited to transaminase elevations (11%) and neutropenia (9%) [342]. These promising results warrant further exploration in phase III trials.

Additional research on ADC combination therapies in hematologic malignancies is summarized in Table 4.

Solid tumors

Although ADC-based combination therapies have shown substantial efficacy in hematologic malignancies, their application in solid tumors is gaining increasing clinical relevance. Substantial evidence suggests that rationally designed ADC combinations, leveraging precision targeting and mechanistic synergy, are broadening the therapeutic landscape for patients with solid tumors.

Breast cancer

BC represents the most intensively studied solid tumor in the field of ADC combination therapy. Among various ADCs, T-DM1 has been the subject of numerous pivotal clinical trials. While combinations of T-DM1 with cytotoxic agents have demonstrated promising efficacy, toxicity remains a major limiting factor. For example, T-DM1 combined with *paclitaxel* achieved an ORR of 50% and a mPFS of 7.4 months in HER2-positive BC, but 90% of patients experienced peripheral neuropathy, and ≥ grade 3 AEs occurred in 77.3% of patients [347]. Likewise, T-DM1 combined with *docetaxel* resulted in an ORR of 80% and a mPFS of 13.8 months, though nearly half of participants required dose reductions due to treatment-related toxicity [348]. These findings underscore the importance of optimizing combination regimens to balance efficacy and tolerability. Several ongoing trials are anticipated to provide further guidance.

Beyond chemotherapy, integrating ADCs with targeted therapies offers additional potential. Given the proven efficacy of dual HER2 blockade, multiple studies have investigated combinations of HER2-targeted ADCs with monoclonal antibodies. The phase III KRISTINE trial compared neoadjuvant T-DM1 plus *pertuzumab* (T-DM1 + P) to *docetaxel/carboplatin/trastuzumab/pertuzumab* (TCH + P) in stage II–III HER2-positive BC. Contrary to expectations, TCH + P achieved significantly higher pathological complete response (pCR) rates (55.7% vs. 44.4%, $P = 0.016$), despite a higher incidence of AEs [349]. At three-year follow-up, T-DM1 + P was associated with increased event-free survival (EFS) events, primarily due to locoregional progression before surgery. Subsequent analysis suggested that patients in the T-DM1 + P arm with lower HER2 expression and increased HER2 heterogeneity were more prone to early progression, indicating that this subgroup may benefit more from conventional chemotherapy combined with HER2 blockade [350].

Similarly, in high-risk early-stage HER2-positive BC, T-DM1 + P achieved a pCR rate of 63% in a phase II trial—lower than that achieved with the THP regimen (*paclitaxel*, *trastuzumab*, *pertuzumab*). Biomarker analysis confirmed strong correlations between HER2 pathway activation and pCR, while luminal A-type tumors demonstrated poorer response [351]. These findings emphasize the need for biomarker-driven treatment individualization to improve therapeutic outcomes.

In advanced HER2-positive BC, ADC combinations continue to demonstrate substantial clinical value. The MARIANNE trial confirmed the non-inferiority of T-DM1 ± P versus *trastuzumab* plus *taxane* (H + T), with mPFS of 13.7, 14.1, and 15.2 months, respectively, and improved safety profiles [352]. More notably, the combination of *tucatinib* and T-DM1 showed enhanced anti-tumor activity in heavily pretreated patients [353],

significantly improving mPFS (9.5 vs. 7.4 months, P = 0.0163), particularly in those with brain metastases [354]. Based on these encouraging results, the phase III HER-2CLIMB-02 trial (NCT03975647) is ongoing. Furthermore, the combination of T-DM1 with *lapatinib* and *nab-paclitaxel* demonstrated favorable tolerance and maintained pharmacokinetic stability [355].

Furthermore, a retrospective study has been conducted evaluating the safety profile of combining ADCs (T-DXd or T-DM1) with radiotherapy in HER2-positive BC with brain metastases. The results demonstrated that although the concurrent ADC-plus-radiotherapy group showed a trend toward improved OS (25.8 vs. 14.1 months, P = 0.092), no significant difference was observed in central nervous system progression-free survival (CNS-PFS) (median, 8.0 vs. 7.8 months, P = 0.86). Notably, the concurrent treatment group exhibited a significantly higher incidence of symptomatic radiation necrosis (SRN), with a 2-year cumulative incidence of 27.4% [356]. Consequently, intracranial safety emerges as a paramount concern requiring careful consideration in combined ADC-radiotherapy strategies.

For HER2-negative BC, Trop-2-targeting ADCs have shown promise. A phase Ib trial evaluating SG combined with *talazoparib* in metastatic TNBC found that sequential rather than concurrent administration yielded superior mPFS (7.6 vs. 2.3 months) and was better tolerated. Notably, BRCA1/2-mutated patients showed higher response rates, offering insight into predictive biomarkers and optimal sequencing strategies [357]. Additional trials on ADC-based combinations in BC are summarized in Table 4 [358–362].

The integration of ADCs with ICIs has also yielded promising, albeit mixed, outcomes. While the KATE2 trial failed to show PFS improvement with *atezolizumab* plus T-DM1 (6.8 vs. 8.2 months, P = 0.33) and revealed increased toxicity [363], other combinations have been more encouraging. For instance, *pembrolizumab* combined with T-DM1 demonstrated a mPFS of 9.6 months and was well tolerated, with no dose-limiting toxicities among 20 enrolled patients [364]. Similarly, the combination of T-DXd with *nivolumab* achieved an ORR of 65.6% in HER2-high and 50% in HER2-low BC [365].

A notable recent advancement was reported in the I-SPY2.2 trial, a multicenter, phase II SMART-design study investigating neoadjuvant therapies in HER2-negative BC. The study design incorporated novel experimental regimens as initial sequential therapy (Block A), followed by standard chemotherapy/targeted therapy (Block B/C) if indicated. The primary objective was to achieve pCR with novel targeted agents alone or in sequential combination with standard therapy, while identifying optimal regimens based on the response predictive subtype (RPS) classification. A total of 106 patients were randomized to the Dato-DXd plus *durvalumab* (Durva) arm. After completing Block A, 35 patients proceeded directly to surgery without requiring Blocks B or C. In the immune + subtype, 43% of patients achieved pCR, demonstrating significant efficacy. When evaluated across the full treatment sequence (Blocks A–C), the overall pCR rate was 50%, with the highest pCR rate (79%) observed in immune + patients. Notably, the high pCR rate in the immune + subgroup suggests that Dato-DXd/*durvalumab* combination may serve as a potential alternative to standard therapy. These findings underscore the need for future immune subtype-guided precision treatments [366]. With upcoming phase III trials [367, 368], we expect to expand therapeutic options for HER2-negative early BC.

Other solid tumors

Although the exploration of ADC-based combination therapies in solid tumors outside of BC has been relatively limited, but several promising results have emerged, demonstrating the potential of these treatments in various malignancies.

In a phase I study involving heavily pretreated patients with platinum-resistant epithelial OC, recurrent endometrial cancer (EC), and TNBC, MIRV combined with gemcitabine was assessed for both safety and efficacy. Preliminary data revealed a PR rate of 45%. However, hematologic AEs remained a significant concern, highlighting the need for careful monitoring of treatment-related toxicity [369]. Similarly, a phase Ib study evaluated *anetumab ravtansine*, a mesothelin-targeting ADC, in combination with *pegylated liposomal doxorubicin* in mesothelin-expressing, platinum-resistant OC. This combination yielded an overall response rate of 27.7%, including 1 complete response and 26.2% PR, with a mPFS of 5.0 months, providing a strong rationale for subsequent phase III trials [370].

Furthermore, the combination of MIRV with *bevacizumab* achieved an ORR of 56% and a mPFS of 9.9 months in patients with *platinum*-resistant OC and high FRα expression [371]. In *platinum*-sensitive populations, the addition of *carboplatin* further increased the response rate to 83%, with a mPFS of 13.5 months. Although this combination was generally well-tolerated, thrombocytopenia remained a key dose-limiting toxicity, underscoring the importance of tailored dosing adjustments based on individual patient characteristics and treatment history [372].

The innovaTV 205/GOG-3024/ENGOT-cx8 trial, which evaluated the tissue factor-targeting ADC *tisotumab vedotin* (TV) in combination with *bevacizumab*, *pembrolizumab*, or *carboplatin* in recurrent or metastatic cervical cancer (r/mCC), reported manageable safety and promising anti-tumor activity in both treatment-naïve

and pretreated patients [373]. These findings suggest that TV-based combinations hold significant potential for improving treatment outcomes in this challenging malignancy.

A notable breakthrough in NSCLC treatment came from the investigation of *telisotuzumab vedotin* (Teliso-V), a c-Met-targeting ADC. When combined with *erlotinib*, Teliso-V demonstrated a mPFS of 5.9 months in c-Met-positive NSCLC patients. Further analysis revealed that in the subgroup of EGFR-mutant patients, those with high c-Met expression achieved an ORR of 52.6%, with T790M-negative patients showing significantly superior mPFS compared to T790M-positive patients (6.8 vs. 3.7 months). These results highlight the importance of monitoring dynamic resistance mutations to guide therapy decisions and optimize treatment outcomes [374]. Additionally, the combination of Teliso-V with osimertinib resulted in an ORR of 50% in EGFR-mutant, c-Met-overexpressing, non-squamous NSCLC patients, demonstrating preliminary efficacy with a manageable safety profile [375].

Beyond BC and NSCLC, substantial progress has also been made in combining ADCs with immunotherapy for solid tumors. EV, when combined with *pembrolizumab*, demonstrated a remarkable ORR of 73.3% and a mOS exceeding 26 months in patients with locally advanced or metastatic UC [376]. These promising results are being further validated in an ongoing phase III clinical trial (NCT04223856). Similarly, the combination of SG with *pembrolizumab* has shown encouraging activity in subsequent-line treatment settings. Preliminary data from the TROPHY-U-01 study revealed a mPFS of 5.3 months and mOS of 12.7 months, with a high response rate and a manageable toxicity profile [377]. These findings represent significant advancements in the treatment of advanced UC, offering new therapeutic options for this challenging disease.

In NSCLC, the TROPION-Lung02 trial evaluated the combination of Dato-DXd with *pembrolizumab*, with or without *platinum*-based chemotherapy. Preliminary results demonstrated ORRs of 60% for the doublet regimen (Dato-DXd + *pembrolizumab*) and 55% for the triplet regimen (Dato-DXd + *pembrolizumab* + *platinum*), with mPFS of 10.8 and 7.8 months, respectively. Although survival data are still immature, these early results suggest that Dato-DXd-based combinations exhibit promising anti-tumor activity with manageable safety in treatment-naïve NSCLC patients [378]. Several ongoing trials are investigating the efficacy of Dato-DXd in combination with immunotherapy, and further results are anticipated in the near future [379, 380].

Moreover, a retrospective study investigated the safety profiles of ADCs combined with radiotherapy across various solid tumors with brain metastases, including BC, NSCLC, esophageal and/or gastric cancer, and salivary gland cancer who received T-DXd or SG with stereotactic radiotherapy (SRT). The results demonstrated that while patients with small brain metastases undergoing initial SRT exhibited manageable risks of SRN, those with larger lesions or concurrent ADC-SRT therapy faced substantially elevated SRN risks [381]. Conversely, recent phase II clinical trial data evaluating HER2-targeted ADC *disitamab vedotin* (DV, a HER2-targeted ADC) combined with *toripalimab* and radiotherapy in localized HER2-positive muscle-invasive bladder cancer (MIBC) revealed favorable tolerability with no grade 4 AEs and durable complete responses observed in 83.3% of patients, establishing a novel bladder-preserving therapeutic paradigm [382]. In summary, the safety profile of ADC-radiotherapy combinations necessitates individualized assessment based on tumor-specific characteristics and radiotherapy target considerations. Future research should focus on optimizing key strategies such as dose adjustment, treatment sequencing, and biomarker-based patient selection to improve therapeutic safety in specific clinical settings.

Additional studies on ADC-based combinations in solid tumors, as summarized in Table 4 [383–396], continue to explore the potential of these therapies in various cancer types.

Challenges and breakthrough strategies

The clinical translation of ADC combination regimens faces numerous challenges, particularly with regard to safety concerns. For example, in a study evaluating the combination of Rova-T with *nivolumab* ± *ipilimumab* in previously treated patients with extensive-stage small cell lung cancer (ES-SCLC), the treatment achieved an ORR of 30%. However, poor tolerance was observed, which significantly limited its clinical applicability [397]. This highlights the critical need for optimized dosing strategies that can strike a balance between efficacy and safety in future combination regimens.

Emerging mechanistic insights have significantly advanced our understanding of how to optimize ADC combinations. Preclinical studies have shown that BV exposure in HL cells induces destabilization of the microtubule network, triggering acute endoplasmic reticulum (ER) stress responses that culminate in ICD. Prolonged BV exposure (72 h) further alters the immune landscape, enhancing the activation of antigen-presenting cells and augmenting the efficacy of PD-1 blockade [398]. These findings provide a compelling preclinical rationale for combining BV with ICIs in clinical settings. Moreover, these data suggest an intriguing possibility: might optimized temporal coordination between ADC and ICI administration, particularly by sequencing immune checkpoint inhibition to align with ICD peaks, enhance

therapeutic efficacy? This hypothesis merits validation in prospective clinical studies.

An additional area of clinical interest is the exploration of dual-ADC combinations, which target distinct antigens and employ different cytotoxic payloads. A notable example is the phase I trial evaluating the combination of SG and EV in metastatic UC, which demonstrated impressive therapeutic synergy. This combination achieved an ORR of 70%, including 3 complete responses [399]. These promising results highlight how rationally designed ADC combinations targeting distinct mechanisms of action may overcome established resistance pathways. However, phase II/III trials remain crucial to confirm the clinical utility of this paradigm-shifting approach.

Conclusions

In this review, we systematically examine the mechanisms underlying resistance to ADCs and evaluate emerging strategies to circumvent these challenges. ADCs have revolutionized cancer therapy by enabling tumor-selective delivery of cytotoxic payloads, yet their clinical efficacy is often hindered by acquired resistance. This resistance is driven by various factors, including target modulation, payload adaptation, and the dynamic remodeling of the TME. Addressing these barriers necessitates a multifaceted approach, as several critical gaps persist. To optimize therapeutic outcomes, it is essential to rigorously define dosing schedules and therapeutic indices in combination regimens, ensuring maximum efficacy while minimizing the risk of synergistic toxicity. However, existing preclinical models often fail to replicate the evolving, patient-specific resistance patterns observed in clinical settings. This disconnect emphasizes the urgent need for functional platforms capable of predicting clonal evolution under therapeutic pressure, thus bridging the gap between bench research and clinical outcomes.

Although clinical trials have laid the groundwork for ADCs development, the field stands to benefit from the widespread adoption of standardized adaptive trial designs. Such designs could facilitate breakthroughs by accelerating the validation of next-generation therapeutic strategies, while also ensuring the rigorous enforcement of safety and efficacy standards. Additionally, integrating biomarker-driven modular treatment regimens is expected to substantially expedite clinical development, especially in the face of complex resistance mechanisms and diverse patient populations. Advances in ADCs engineering from optimized linker chemistries to bispecific targeting and synergistic combinations are collectively strengthening our therapeutic capabilities. Moreover, emerging technologies, such as liquid biopsy and multiomics analysis, enable real-time monitoring of resistance mechanisms, offering new opportunities to advance precision oncology.

Looking to the future, unlocking the full potential of ADCs will require overcoming current obstacles through interdisciplinary innovation. This includes the enhancement of tumor-specific, stimulus-responsive delivery systems, the optimization of drug pairings via AI, and the further refinement of ADC architectures based on synthetic biology principles. When coupled with biomarker-driven adaptive trials, these innovations are poised to significantly advance the clinical application of ADCs across a wide range of malignancies, transforming cancer treatment from incremental improvements to sustained, personalized therapies.

Notably, at the recent American Society of Clinical Oncology (ASCO) Annual Meeting, several ADC-related studies presented promising clinical data. For example, DV in combination with *toripalimab* (Tor) and chemotherapy (C) or *trastuzumab* (Tra) demonstrated substantial clinical benefits in the first-line treatment of HER2-positive locally advanced or metastatic GC. The latest data indicate that the DV + Tor + Tra combination significantly improved the ORR, highlighting its potential as a chemotherapy-free treatment option. In patients with HER2-low gastric cancer and gastroesophageal junction cancer, the DV + Tor + CAPOX combination therapy showed superior ORR and PFS, with favorable safety profiles [400].

Additionally, the novel Nectin-4-targeted ADC *9MW2821*, in combination with toripalimab, demonstrated an ORR of 87.5% and DCR of 92.5% in first-line treatment of patients with locally advanced or metastatic UC, underscoring its potential efficacy across various subgroups. Notably, the treatment response was particularly striking in patients with liver metastases and those with Nectin-4-negative tumors, suggesting that this combination therapy may benefit different patient populations [401]. In SCLC, *BL-B01D1* exhibited promising efficacy, with an ORR of 55.2%, a mPFS of 4.0 months, and a mOS of 12.0 months, further demonstrating its potential as a new treatment option for SCLC [402]. Similarly, *IBI343*, the first ADC targeting CLDN18.2, showed promising results in advanced PDAC, particularly in CLDN18.2-positive patients, with an ORR of 22.7%, a DCR of 81.8%, and a mPFS of 5.4 months [403]. Furthermore, SHR-1826, an ADC targeting MET alterations, displayed significant anti-cancer activity in advanced solid tumors, especially in NSCLC patients, with an ORR of 39.7% and a DCR of 94.8%, providing new therapeutic hope for this patient group [404].

These clinical findings, along with other relevant studies, further underscore the substantial potential of ADCs in anti-tumor therapy [405–408]. They also highlight the shift of ADC-based combination therapies from simple

additive effects to more sophisticated, multi-faceted strategies, reflecting the continued progress in this field. With ongoing optimization of therapeutic strategies and the integration of cutting-edge technologies, ADCs are expected to play an increasingly prominent role in cancer treatment.

In conclusion, realizing the full potential of ADCs will depend on overcoming current challenges through a convergence of innovations: stimuli-responsive delivery systems to enhance tumor specificity, AI-driven drug pairing to optimize combination regimens, and synthetic biology-inspired engineering to refine ADC architectures. When combined with biomarker-adaptive trials, these advances hold the promise of revolutionizing ADC therapy across diverse malignancies, advancing cancer treatment from incremental progress to durable, patient-tailored therapies.

Abbreviations

ICIs	Immune checkpoint inhibitors
ADCs	Antibody-drug conjugates
ICD	Immunogenic cell death
GO	Gemtuzumab ozogamicin
AML	Acute myeloid leukemia
BV	Brentuximab vedotin
HL	Hodgkin lymphoma
sALCL	Systemic anaplastic large cell lymphoma
T- DM1	Trastuzumab emtansine
BC	Breast cancer
TME	Tumor microenvironment
P- gp	P-glycoprotein
MRPs	Multidrug resistance-associated proteins
BCRP	Breast cancer resistance protein
MDR	Multidrug resistance
ABC	ATP-binding cassette
MMAE	Monomethyl auristatin E
Dato-DXd	Datopotamab deruxtecan
T- DM1	Trastuzumab emtansine
SN-38	7-ethyl-10-hydroxycamptothecin
BBB	Blood-brain barrier
siRNA	Small interfering RNA
GSH	Glutathione
UC	Urothelial carcinoma
EV	Enfortumab vedotin
PDAC	Pancreatic ductal adenocarcinoma
GC	Gastric cancer
ccRCC	Clear cell renal cell carcinoma
BITC	Benzyl isothiocyanate
CSA	Cyclosporine A
BSO	Buthionine sulfoximine
ABCG2	ATP-binding cassette sub-family G member 2
PDT	Photodynamic therapy
PpIX	Pprotoporphyrin IX
miRNAs	microRNAs
3'-UTR,	3'-untranslated region
SG	Sacituzumab govitecan
TNBC	Triple-negative breast cancer
TROP2	Trophoblast cell-surface antigen 2
IO	Inotuzumab ozogamicin
B- ALL	B-cell acute lymphoblastic leukemia
CR	Complete response
PFS	Progression-free survival
Endo A2	Endophilin A2
CAV-1	Caveolin-1
scRNA-seq	Single-cell RNA sequencing
TKIs	Tyrosine kinase inhibitors
DDR	DNA damage response
Treg	Regulatory T cell
CAF	Cancer-associated fibroblast
ctDNA	Circulating tumor DNA
IFP	Interstitial fluid pressure
ECM	Extracellular matrix
HA	Hyaluronan
EMT	Epithelial-mesenchymal transition
PEGPH20	PEGylated hyaluronidase
HCC	Hepatocellular carcinoma
CNN1	Calponin 1
FAP	Fibroblast activation protein
TAMs	Tumor-associated macrophages
FcγR	Fc–Fcγ receptor
mAb	Monoclonal antibody
APCs	Antigen-presenting cells
Rova-T	Rovalpituzumab tesirine
SCLC	Small-cell lung cancer
MDSC	Myeloid-derived suppressor cell
PDK	Pyruvate dehydrogenase kinase
OS	Overall survival
NSCLC	Non-small cell lung cancer
ORR	Objective response rate
DCR	Disease control rate
PR	Partial response
HR	Hazard ratio
OC	Ovarian cancer
MIRV	Mirvetuximab soravtansine
CNS	Central nervous system
CTCs	Circulating tumor cells
DAC	Decitabine
CTAs	Cancer-testis antigens
HNSCC	Head and neck squamous cell carcinoma
DLBCL	Diffuse large B-cell lymphoma
EZH2	Enhancer of zeste homolog 2
PRC2	Polycomb repressive complex 2
HDACIs	Histone deacetylase inhibitors
HNRNPU	Heterogeneous nuclear ribonucleoprotein U
ADCC	Antibody-dependent cellular cytotoxicity
bsADCs	Bispecific ADCs
PDX	Patient-derived xenograft
Bcl-xL	B-cell lymphoma-extra large
ABBV-155	Mirzotamab cletuzoclax
DAR	Drug-to-antibody ratio
NMR	Nuclear magnetic resonance
ISACs	Immune-stimulating antibody conjugates
TLR	Toll-like receptor
STING	Stimulator of interferon genes
bsAb	Bispecific antibody
SORT1	Sortilin-1
NIR	Near-infrared
ROS	Reactive oxygen species
GPC1	Glypican-1
VEGF	Vascular endothelial growth factor
ROCK	Rho-associated protein kinase
MLC	Myosin light chain
LIMK	LIM kinase
CRS	Cytokine release syndrome
B-NHL	B-cell non-Hodgkin lymphoma
PV	Polatuzumab vedotin
AEs	Adverse events
pCR	Pathological complete response
EFS	Event-free survival
CNS-PFS	Central nervous system progression-free survival
SRN	Symptomatic radiation necrosis
RPS	Response predictive subtype
EC	Endometrial cancer
TV	Tisotumab vedotin
r/mCC	Recurrent or metastatic cervical cancer
Teliso-V	Telisotuzumab vedotin
SRT	Stereotactic radiotherapy
DV	Disitamab vedotin

MIBC	Muscle-invasive bladder cancer
ES-SCLC	Extensive-stage small cell lung cancer
ER	Endoplasmic reticulum
AI	Artificial intelligence
ASCO	American Society of Clinical Oncology

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